
UNIFYING FORMAL EXPLANATIONS: A COMPLEXITY-THEORETIC PERSPECTIVE

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ABSTRACT

The computational complexity of deriving two canonical families of explanations for ML model predictions has been the subject of prior work: (i) *sufficient reasons*, defined as subsets of input features whose fixing determines the prediction, and (ii) *contrastive reasons*, defined as subsets of input features whose perturbation alters the prediction. Existing analyses treat these explanation forms under disparate specifications — non-probabilistic versus probabilistic formulations, and local versus global scopes. In this work, we formulate a unified framework in which every such explanation form is characterized as the minimization of a single probabilistic value function subject to a threshold constraint. We then establish that the computational complexity of this minimization is governed by three structural properties of the value function: (i) *monotonicity*, (ii) *submodularity*, and (iii) *supermodularity* — three properties of central importance in *combinatorial optimization*. Our analysis yields several counterintuitive characterizations of how these properties are instantiated across the explanation settings under study. Specifically, whereas the *local* value functions satisfy none of monotonicity, submodularity, or supermodularity, we prove that the *global* value functions provably satisfy them. This structural dichotomy is what permits us to derive a series of new polynomial-time computability results for explanations with provable guarantees in the global setting, spanning a range of ML models that traverse the interpretability spectrum, including neural networks, decision trees, and tree ensembles. Conversely, we prove that even substantially relaxed instances of these explanation problems are NP-hard in the corresponding local setting.

1 INTRODUCTION

Despite substantial methodological progress in explaining ML model decisions, a recurring finding in the literature is that many explanation types carrying provable guarantees are computationally intractable to compute (Barceló et al., 2020; Van den Broeck et al., 2022), and the intractability tends to intensify as model expressiveness increases, i.e., for complex or highly non-linear models (Barceló et al., 2020; Adolfi et al., 2025; 2024). Consequently, the *computational complexity* of computing explanations has emerged as a central theoretical concern, and a substantial body of recent work seeks to delineate precisely which explanation types admit efficient computation for which model classes, and which are provably out of reach (Barceló et al., 2020; Wäldchen et al., 2021; Arenas et al., 2022; 2023; Marzouk & De La Higuera, 2024; Ordyniak et al., 2023; Laber, 2024; Bhattacharjee & Luxburg, 2024; Blanc et al., 2021; 2022).

From sufficient to contrastive reasons. Among studies on the computational complexity of generating explanations, two fundamental types of explanations for ML models were extensively examined: (i) *sufficient reasons* and (ii) *contrastive reasons* (Barceló et al., 2020; Arenas et al., 2022; Audemard et al., 2022a; Arenas et al., 2021; Barceló et al., 2025; Marques-Silva & Ignatiev, 2022; Ignatiev et al., 2020b; Darwiche & Hirth, 2020). A *sufficient reason* is a feature subset S such that fixing the coordinates in S to prescribed values determines the model prediction: the prediction is invariant to every assignment of the complementary coordinates $\bar{S}$. A *contrastive reason* is, dually, a feature subset S such that some perturbation of the coordinates in S induces a change in the model prediction.

In contrast to additive attribution methods, which assign real-valued importance scores to individual features but are frequently difficult for humans to interpret (Kumar et al., 2020) and lack actionabil-

ity (Bilodeau et al., 2024), sufficient and contrastive reasons yield *discrete, condition-like explanations* that answer directly the decision-theoretic questions “which features suffice to certify this prediction?” and “which features must be altered to invert it?”. This discrete semantics accounts for their central role in many established XAI methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dhurandhar et al., 2018), for their demonstrated superiority over additive methods in improving human predictive accuracy (Ribeiro et al., 2018; Yin & Neubig, 2022; Dhurandhar et al., 2018), and for their utility in downstream tasks including bias detection (Balkir et al., 2022; Carter et al., 2021; La Malfa et al., 2021; Muthukumar et al., 2018), model debugging (Jacovi et al., 2021), and anomaly detection (Davidson et al., 2025). A further well-established principle holds that sufficient and contrastive explanations of *smaller cardinality* are more interpretable, which elevates *minimality* to a central formal guarantee of interest (Ribeiro et al., 2018; Lopardo et al., 2023; Carter et al., 2019; Barceló et al., 2020; Arenas et al., 2022; Blanc et al., 2021; Wäldchen et al., 2021).

Barceló et al. (2020) conducted one of the earliest complexity-theoretic analyses of computing sufficient and contrastive reasons. They established that computing a cardinally minimal sufficient reason is NP-hard for decision trees, and that its complexity rises to Σ_2^P -hard for neural networks. For contrastive reasons, they showed that computing a cardinally minimal explanation is solvable in polynomial time for decision trees, whereas the same task is NP-hard for neural networks. Analogous hardness results were subsequently established for tree ensembles (Izza & Marques-Silva, 2021; Ordyniak et al., 2024; Audemard et al., 2022b).

From non-probabilistic to probabilistic explanations. A recurring criticism of the classical definitions of sufficient and contrastive reasons is that their guarantees are enforced in an absolute, worst-case sense over the entire input domain, which renders them rigid and, in turn, prone to producing explanations of excessive cardinality or low informativeness (Ignatiev et al., 2019; 2020a; Arenas et al., 2022; Wäldchen et al., 2021). To relax this worst-case requirement, the literature has adopted a more general, *probabilistic* formulation of sufficient and contrastive reasons (Wäldchen et al., 2021; Arenas et al., 2022; Blanc et al., 2021; Izza et al., 2023; Xue et al., 2023). In this formulation, the objective is to identify feature subsets that determine or alter the prediction with probability at least a prescribed threshold δ , measured with respect to an input distribution.

Wäldchen et al. (2021) were the first to characterize the complexity of probabilistic sufficient reasons, proving NP^{PP} -hardness for CNF classifiers. This lower bound propagates to tree ensembles and neural networks (Barceló et al., 2020; Ordyniak et al., 2023). A pivotal structural observation in the probabilistic regime is that the underlying probability function *fails to be monotone* (Arenas et al., 2022; Izza et al., 2023), and it is precisely this failure that renders even *subset-minimal* explanations computationally hard. Notably, (Arenas et al., 2022) prove that computing a subset-minimal probabilistic sufficient reason is NP-hard already for decision trees — in sharp contrast to the non-probabilistic setting, in which the analogous explanation is computable in polynomial time (Huang et al., 2021; Barceló et al., 2020).

From local to global explanations. More recently, Bassan et al. (2024) extended the complexity analysis from the *local* (non-probabilistic) setting — in which an explanation is bound to a single prediction — to the *global* (non-probabilistic) setting, which requires sufficient or contrastive reasons that hold uniformly over the entire input domain. As with other non-probabilistic formulations (Ignatiev et al., 2019; Barceló et al., 2020; Arenas et al., 2021; Darwiche & Hirth, 2020), however, the resulting criteria are highly stringent — arguably more so globally than locally. Concretely, any feature omitted from a global sufficient reason is certified to be strictly redundant (Bassan et al., 2024), which typically forces the explanation to encompass nearly all input coordinates and thereby degrades its informativeness.

OUR CONTRIBUTIONS

1. We subsume the previously disparate explanation-computation problems — sufficient and contrastive, probabilistic and non-probabilistic, local and global — within a single framework, cast as the minimization of a unified value function subject to a threshold constraint. Within this framework we isolate three structural properties of the value function that provably govern the complexity of the task: (a) *monotonicity*, (b) *supermodularity*, and (c) *submodularity*.

2. We establish that these properties are instantiated in *qualitatively different ways* depending on the structure of the value function. In particular, we prove the counterintuitive result that, although the *local* value functions for both sufficient and contrastive reasons are *non-monotone*, their *global* counterparts are *monotone non-decreasing*. We further identify additional structural properties specific to the global setting: the global sufficient value function is *supermodular*, whereas the global contrastive value function is *submodular* — neither of which holds in the local setting.
3. We exploit these properties to derive new complexity results for explanation computation, establishing that global explanations with provable guarantees admit efficient computation even where their local counterparts are computationally hard. We instantiate these results across three widely used model classes spanning the interpretability spectrum: (i) neural networks, (ii) decision trees, and (iii) tree ensembles. First, we prove that although computing a subset-minimal local probabilistic sufficient/contrastive explanation is NP-hard already for decision trees (Arenas et al., 2022), the corresponding global problem is solvable in polynomial time. We then extend this positive result to arbitrary black-box models (including complex models such as neural networks and tree ensembles) under empirical distributions: computing a subset-minimal global sufficient/contrastive explanation is achievable in polynomial time, whereas the local problem remains NP-hard for these models.
4. Finally, we establish a stronger complexity separation by exploiting the submodular and supermodular structure of the global value functions — structure that is absent in the local case. Concretely, we prove that provable *constant-factor* approximation guarantees are attainable for computing cardinally minimal global explanations — even for complex models such as neural networks and tree ensembles — under a fixed empirical distribution. In sharp contrast, we establish strong inapproximability lower bounds in the local setting, showing that no *bounded-factor* approximation exists even under highly simplified assumptions.

Owing to space constraints, the main text states our principal theorems and corollaries and outlines the underlying arguments, while the complete proofs are deferred to the appendix.

2 PRELIMINARIES

Setting. We fix an input dimension $n \in \mathbb{N}$ and denote input vectors by $\mathbf{x} := (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$. Each coordinate $\mathbf{x}_i$ ranges over a corresponding domain $\mathcal{X}_i$, which may be discrete or continuous, so that the input feature space is the product $\mathbb{F} := \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n$. We study classification models $f : \mathbb{F} \rightarrow [c]$, where $c \in \mathbb{N}$ denotes the number of classes, together with a *generic* input distribution $\mathcal{D} : \mathbb{F} \rightarrow [0, 1]$. Because accurate approximation of $\mathcal{D}$ is computationally infeasible in many settings, a standard alternative is to substitute a fixed empirical dataset $\mathbf{D} := \{\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}\}$ as a tractable proxy for the underlying distribution — for instance, by taking $\mathbf{D}$ to be a sample drawn from the available training data.

The explanations we study are either local or global. A *local* explanation is defined relative to a specific input $\mathbf{x} \in \mathbb{F}$ and accounts for the prediction $f(\mathbf{x})$ at that instance. A *global* explanation, by contrast, is defined independently of any particular $\mathbf{x}$ and characterizes the decision behavior of f across the input space as a whole.

Models. While many of our results are model-agnostic (e.g., intrinsic properties of the value functions), we additionally establish model-specific complexity results for widely used ML model classes. To cover the interpretability spectrum, we analyze model classes ranging from those conventionally regarded as “black-box” to those regarded as “interpretable”. Concretely, we consider: (i) decision trees, (ii) neural networks (encompassing all architectures at least as expressive as feed-forward ReLU networks), and (iii) tree ensembles, including majority-voting ensembles (e.g., random forests) and weighted-voting ensembles (e.g., XGBoost). The corresponding formal definitions are deferred to the Appendix.

Distributions. In the probabilistic explanation settings, the complexity depends critically on the input distribution $\mathcal{D}$. We therefore distinguish three distribution classes: (i) *general distributions*, which impose no assumptions on $\mathcal{D}$ and hence subsume all distributions; we invoke this class primarily when establishing properties that hold universally; (ii) *empirical distributions*, comprising all distributions induced by the finite dataset $\mathbf{D}$ — a standard construction in XAI (Lundberg & Lee, 2017; Van den

Broeck et al., 2022); and (iii) *independent distributions*, which posit mutual independence of the features under $\mathcal{D}$ — another widely adopted assumption in the XAI literature (Arenas et al., 2022; 2023; Lundberg & Lee, 2017; Ribeiro et al., 2018). We stress that empirical distributions do not entail feature independence; on the contrary, they can encode complex feature dependencies extracted from finite datasets (Van den Broeck et al., 2022). The full formal definitions of all three classes are provided in the Appendix.

3 FORMS OF REASONS

This section formalizes the explanation types studied in this work. We begin with the strict non-probabilistic definitions of (local and global) sufficient and contrastive reasons, and then generalize them to their probabilistic counterparts, which recover the non-probabilistic definitions as the special case $\delta = 1$.

3.1 Non-Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

Sufficient reasons. A common goal of *feature selection* is to isolate a small set of features that account for a model’s decision. We adopt the well-established *sufficiency* criterion for this purpose, which underlies many widely used explainability methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dasgupta et al., 2022), and which may be imposed either locally — with respect to a single prediction — or globally — over the entire input domain. Following standard conventions, we define a *local sufficient reason* as a feature subset $S \subseteq \{1, \dots, n\}$ such that fixing the coordinates in S to their values in $\mathbf{x}$ pins the prediction to $f(\mathbf{x})$ for every assignment of the complementary coordinates $\bar{S}$. Formally, S is a local sufficient reason for $\langle f, \mathbf{x} \rangle$ iff:

$$\forall \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (1)$$

Here $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ denotes the vector whose coordinates in S are taken from $\mathbf{x}$ and whose coordinates in $\bar{S}$ are taken from $\mathbf{z}$. Local sufficient reasons are closely connected to *semi-factual* explanations (Kenny & Keane, 2021; Alfano et al., 2025), which seek an alternative input $\mathbf{z}'$ that preserves the prediction (i.e., $f(\mathbf{z}') = f(\mathbf{x})$) while minimizing its distance to the original point. Instantiating $\mathbf{z}'$ as $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ and taking the Hamming distance as the proximity measure makes the two notions coincide.

A *global sufficient reason* (Bassan et al., 2024) is a feature subset $S \subseteq \{1, \dots, n\}$ that is simultaneously a local sufficient reason for *every* input $\mathbf{x} \in \mathbb{F}$:

$$\forall \mathbf{x}, \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (2)$$

Contrastive reasons. A complementary approach to explanation identifies feature subsets whose perturbation *alters* a prediction (Dhurandhar et al., 2018; Mothilal et al., 2020; Guidotti, 2024), thereby characterizing the minimal changes required to flip the model output. Formally, a subset $S \subseteq \{1, \dots, n\}$ is a *local contrastive reason* for $\langle f, \mathbf{x} \rangle$ iff:

$$\exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (3)$$

Contrastive reasons are, dually, closely connected to *counterfactual* explanations (Guidotti, 2024; Mothilal et al., 2020), which seek a nearby assignment $\mathbf{x}'$ at which the prediction flips, i.e., $f(\mathbf{x}') \neq f(\mathbf{x})$. Taking the distance metric to be the Hamming weight again makes the two notions coincide.

In direct analogy with the sufficient case, one may require a feature subset whose perturbation alters the prediction for *every* input in the domain. This global form of explanation is closely connected to bias-detection methods (Arenas et al., 2021; Bassan et al., 2024; Darwiche & Hirth, 2020), as well as to *group* counterfactual methods that compute counterfactuals jointly over multiple data instances (Carrizosa et al., 2024; Warren et al., 2024). Formally, S is a *global contrastive reason* for f iff:

$$\forall \mathbf{x} \in \mathbb{F}, \quad \exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (4)$$

3.2 Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

As noted in the introduction, the non-probabilistic definitions impose worst-case conditions that are often overly restrictive, which motivates their relaxation to *probabilistic* definitions (Arenas et al., 2022; Ribeiro et al., 2018; Izza et al., 2023; Wäldchen et al., 2021; Bounia & Koriche, 2023; Subercaseaux et al., 2025; Wang et al., 2021a; Blanc et al., 2021). These definitions replace the universal quantifier with the weaker requirement that the guarantee hold with probability at least $\delta \in [0, 1]$ under $\mathcal{D}$; setting $\delta = 1$ recovers the non-probabilistic definitions exactly.

Probabilistic sufficient reasons. We say that $S \subseteq \{1, \dots, n\}$ is a *local δ -sufficient reason* for $\langle f, \mathbf{x} \rangle$ if, conditioning on the coordinates in S being fixed to their values in $\mathbf{x}$ and sampling the complementary coordinates from $\mathcal{D}$, the prediction is preserved with probability at least δ ; that is:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \geq \delta. \quad (5)$$

where the conditioning event $\mathbf{z}_S = \mathbf{x}_S$ fixes the coordinates of $\mathbf{z}$ indexed by S to their values in $\mathbf{x}$. Following the standard construction of global explanations as an average over the input domain, we define $S \subseteq \{1, \dots, n\}$ to be a *global δ -sufficient reason* for $\langle f \rangle$ if the expectation of the local sufficiency probability, taken over $\mathbf{x}$ drawn from $\mathcal{D}$, is at least δ ; that is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \geq \delta. \quad (6)$$

Probabilistic contrastive reasons. Paralleling the non-probabilistic case, a *local δ -contrastive reason* is a feature subset whose perturbation alters the prediction with sufficiently high probability. In contrast to the sufficient case, here we condition on the *complementary* coordinates $\bar{S}$ being fixed to their values in $\mathbf{x}$, and, letting the coordinates in S vary under $\mathcal{D}$, we require the prediction to differ from $f(\mathbf{x})$ with probability at least δ . Formally:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}) \geq \delta. \quad (7)$$

For the global setting, and in direct analogy with global sufficient reasons, we define S to be a *global δ -contrastive reason* if the expectation of the local contrastive probability, taken over $\mathbf{x}$ drawn from $\mathcal{D}$, is at least δ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (8)$$

4 FORMS OF MINIMALITY

As noted in the introduction, across all explanation forms introduced above — sufficient or contrastive, local or global — the literature standardly posits that explanations of *smaller cardinality* are more informative, which makes *minimality* a provable guarantee of primary interest. We formalize two distinct notions of minimality, both applicable to every explanation form we consider:

Definition 1. Assuming a subset $S \subseteq \{1, \dots, n\}$ is an explanation, then:

1. S is a *cardinally-minimal explanation* (Barceló et al., 2020; Bassan et al., 2024) iff S has the smallest explanation cardinality $|S|$ (i.e., there is no explanation S' such that $|S'| < |S|$).
2. $S \subseteq \{1, \dots, n\}$ is a *subset-minimal explanation* (Arenas et al., 2022; Ignatiev et al., 2019) iff S is an explanation, and any $S' \subsetneq S$ is not an explanation.

Cardinal minimality is strictly stronger than subset minimality: every cardinally minimal S is subset-minimal, whereas the converse fails (Appendix B.3 provides an explicit counterexample). We use the terminology “subset-minimal” and “cardinally minimal”, rather than “local” and “global” minima, to avoid conflation with the local/global explanation distinction (input-level versus domain-level reasoning). Both notions instantiate on *every* explanation form under study; for instance, a cardinally minimal local probabilistic δ -sufficient reason is one of minimum cardinality $|S|$, whereas a subset-minimal one is any qualifying S admitting no qualifying proper subset $S' \subsetneq S$.

5 A UNIFIED COMBINATORIAL OPTIMIZATION TASK

A key observation is that every computational problem introduced above — local or global, sufficient or contrastive, probabilistic or non-probabilistic — reduces to computing a minimum-cardinality subset S subject to the constraint $v(S) \geq \delta$, where the non-probabilistic instances correspond to setting $\delta := 1$. We fix the following notation for the four value functions: v_{suff}^ℓ denotes the local sufficiency probability of equation 5, i.e., $\Pr_{\mathbf{z} \sim \mathcal{D}_p}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$, and v_{suff}^g its global counterpart from equation 6; likewise, v_{con}^ℓ and v_{con}^g denote the local and global contrastive probabilities from equations 7 and 73, respectively. With this notation in place, we state the unified problem of computing a cardinally minimal local/global δ -sufficient/contrastive reason:

Cardinally Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and $|S|$ is minimal.

Relaxing the minimality requirement from cardinal to subset minimality yields the following weaker optimization objective:

Subset Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and for any $S' \subseteq S$ it holds that $v(S') < \delta$.

Properties of v governing the complexity. We now isolate the structural properties of the value function that, as we establish subsequently, determine the complexity of explanation computation. The first is *non-decreasing monotonicity*, equivalent to the requirement that every marginal contribution $v(S \cup \{i\}) - v(S)$ be non-negative. Formally:

Definition 2. We say that a value function v maintains non-decreasing monotonicity if for any $S \subseteq \{1, \dots, n\}$ and any $i \in \{1, \dots, n\}$ it holds that: $v(S \cup \{i\}) \geq v(S)$.

The remaining properties, *supermodularity* and *submodularity*, characterize how the marginal contribution $v(S \cup \{i\}) - v(S)$ evolves as the conditioning set grows. In the supermodular case, this marginal contribution is non-decreasing in S (marginal returns increase with set inclusion); dually, in the submodular case, it is non-increasing in S (marginal returns diminish). Formally:

Definition 3. Let there be some value function v , some $S \subseteq S' \subseteq \{1, \dots, n\}$, and $i \notin S'$. Then:

1. v maintains supermodularity iff it holds that: $v(S \cup \{i\}) - v(S) \leq v(S' \cup \{i\}) - v(S')$.
2. v maintains submodularity iff it holds that: $v(S \cup \{i\}) - v(S) \geq v(S' \cup \{i\}) - v(S')$.

6 UNRAVELING NEW PROPERTIES OF THE GLOBAL VALUE FUNCTIONS

Prior work establishes that, in the local non-probabilistic setting, the tractability of computing subset-minimal sufficient or contrastive reasons rests on monotonicity of the value function — a property that holds only in the restricted regime $\delta = 1$. This regime is severely limiting and offers no probabilistic flexibility. One might hope to recover monotonicity under probabilistic guarantees for general δ ; however, prior results show that monotonicity fails in this regime, which in turn renders explanation computation harder (Arenas et al., 2022; Kozachinskiy, 2023; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b).

In the still more restrictive *global non-probabilistic* setting, Bassan et al. (2024) established a stringent *uniqueness* property: there exists exactly *one* subset-minimal explanation. The requirement $\delta = 1$ makes this setting highly restrictive — more so than the local case, since the sufficiency condition must now hold for *all* inputs simultaneously. Indeed, Bassan et al. (2024) shows that this unique minimal subset coincides with the set of all features that are not strictly redundant, which in practice may be the entire feature set. We prove that in the general global *probabilistic* setting — for arbitrary δ — this uniqueness property fails, and the number of subset-minimal explanations may be *exponential* in n .

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Although uniqueness thus holds only in the special case $\delta = 1$ and fails for general δ , we show that the essential *monotonicity* property does hold for the global value functions across *all* values of δ — for both the sufficient (v_{suff}^g) and the contrastive (v_{con}^g) value functions. This stands in sharp contrast to the local setting, in which the corresponding value functions v_{suff}^ℓ and v_{con}^ℓ are non-monotone:

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Beyond the dichotomy that monotonicity holds globally but fails locally, we identify further structural properties specific to the global setting, namely submodularity and supermodularity. Concretely, under the standard assumption of feature independence, the global sufficient value function v_{suff}^g is supermodular, whereas its local counterpart v_{suff}^ℓ fails to be supermodular even under the strictly stronger assumption of a uniform (hence independent) input distribution. Specifically:

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.*

For the second family of settings — computing a global probabilistic *contrastive* reason — we show that the corresponding value function is not supermodular but rather *submodular*, the dual property. This again contrasts with the local setting, in which the value function is neither submodular nor supermodular.

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^ℓ is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

7 COMPUTATIONAL COMPLEXITY RESULTS

7.1 SUBSET MINIMAL EXPLANATIONS

This section analyzes the complexity of computing *subset-minimal* explanations (local or global, sufficient or contrastive) across the model classes under study. The decisive property is *monotonicity* of the value function. As established in the previous section, monotonicity holds for both global value functions v_{con}^g and v_{suff}^g but fails for the local value functions v_{con}^ℓ and v_{suff}^ℓ . This dichotomy is precisely what guarantees that a greedy elimination procedure converges to a subset-minimal explanation in the global setting yet may fail to do so locally. Consequently, we establish that computing subset-minimal explanations is intractable in the local setting but polynomial-time solvable in the global setting. We first introduce the following generalized greedy procedure:

Algorithm 1 Subset Minimal Explanation Search

Input Value function v , and some $\delta \in [0, 1]$

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1:  $S \leftarrow \{1, \dots, n\}$ 
2: while  $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \setminus \{i\})$ 
4:    $S \leftarrow S \setminus \{j\}$ 
5: end while
6: return  $S$ 

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$\triangleright S$ is a (subset minimal?) δ -explanation

Algorithm 1 computes a candidate subset-minimal δ -explanation by top-down greedy elimination. It initializes S to the full feature set $\{1, \dots, n\}$ and, at each iteration, tests whether the largest attainable value $\max_i v(S \setminus \{i\})$ still meets the threshold δ ; if so, it removes the feature j whose deletion incurs

the smallest loss in value, i.e., $j \leftarrow \operatorname{argmax}_{i \in S} v(S \setminus \{i\})$. The procedure terminates and returns S once no single feature can be removed without violating the constraint.

Whether Algorithm 1 returns a genuinely subset-minimal explanation is determined by the *monotonicity* of v . At termination, removing any single feature i from S drives $v(S \setminus \{i\})$ below δ ; monotonicity strengthens this local termination condition into the required guarantee that $v(S \setminus S') < \delta$ for every nonempty $S' \subseteq S$. Combining this with the monotonicity results established in the previous sections yields the following proposition:

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Building on this convergence result, we now establish new complexity results for computing the various forms of subset-minimal explanations within our framework, instantiated for each of the model classes under study.

Decision trees. We first treat a deliberately simplified, ostensibly “interpretable” regime: the model f is a decision tree and the distribution $\mathcal{D}$ is independent. Even here, we exhibit a strict local/global separation. Arenas et al. (2022) proved the intractability result that, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm computes a subset-minimal local δ -sufficient reason for decision trees under independent distributions (and, in fact, already under the uniform distribution). We prove, conversely, that the very same problem is tractable in the global setting: a subset-minimal global δ -sufficient reason for decision trees is computable in polynomial time. Formally:

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Extension to other tractable models. An analogous tractability guarantee for the global setting extends to *orthogonal DNFs* (Crama & Hammer, 2011) — Disjunctive Normal Form formulas whose terms are pairwise mutually exclusive. This extension is of interest because orthogonal DNFs *strictly generalize* decision trees while retaining the structural properties our argument exploits, so the guarantee carries beyond tree-structured models. The full argument is deferred to Appendix N.

Neural networks, tree ensembles, and other complex models. We now lift these results to model classes more expressive than decision trees. We show that, whenever $\mathcal{D}$ is an empirical distribution and the model f admits polynomial-time inference — a mild and standard assumption — computing a subset-minimal global δ -sufficient or δ -contrastive reason is achievable in polynomial time.

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

This positive result, which holds for *any* model under the empirical-distribution assumption, sharpens the separation between the local and global settings. Indeed, for sufficiently expressive models, computing subset-minimal *local* explanations remains intractable even under empirical distributions. We establish this intractability for both neural networks and tree ensembles, obtaining the following strict complexity separation:

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $\text{PTIME}=\text{NP}$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

7.2 APPROXIMATE CARDINALLY MINIMAL EXPLANATIONS

We now turn from subset-minimal reasons to the harder task of computing a *cardinally minimal* δ -sufficient/contrastive reason. We show that, in the global setting, the joint structure of monotonicity together with supermodularity or submodularity yields *provable approximation guarantees* for this task, whereas the local counterpart remains intractable. This further reinforces the separation between the tractability of global and local explanation computation.

A unified greedy approximation algorithm. For a monotone non-decreasing submodular objective, computing a cardinally minimal explanation instantiates the *submodular set cover* problem (Wolsey, 1982); for a *supermodular* objective, it instantiates a non-submodular variant of the same problem (Shi et al., 2021). Both problems are well studied and admit strong approximation guarantees (Wolsey, 1982; Iyer & Bilmes, 2013; Chen & Crawford, 2023). The standard algorithmic device for submodular set cover is the classical greedy algorithm, which we state next (Algorithm 2); it is the common basis for approximating both cardinally minimal sufficient and cardinally minimal contrastive δ -reasons, and we analyze its guarantee in each case below.

Algorithm 2 Cardinally Minimal Explanation Approximation Search

Input Value function v , and some $\delta \in [0, 1]$

```

1:  $S \leftarrow \emptyset$ 
2: while  $\max_{i \in S} v(S \cup \{i\}) < \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \cup \{i\})$ 
4:    $S \leftarrow S \cup \{j\}$ 
5: end while
6: return  $S$   $\triangleright |S|$  is a (provable approximation?) of a cardinally minimal  $\delta$ -explanation

```

Algorithm 2 is the bottom-up dual of Algorithm 1: it initializes S to the empty set and greedily inserts features, at each step choosing the feature $j \leftarrow \operatorname{argmax}_i v(S \cup \{i\})$ that maximizes the value, and terminating once the threshold δ is met.

Cardinally minimal contrastive reasons. Consider first global contrastive reasons. Here monotonicity and submodularity both hold, so the task reduces exactly to *submodular set cover*, and Wolsey (1982)’s classical guarantee applies to Algorithm 2. For integer-valued objectives the algorithm attains a Harmonic-number factor, and more generally an $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation. Under an empirical distribution $\mathcal{D}$ of fixed sample size, this factor collapses to a *constant* that grows only *logarithmically* in the sample size, a substantially strong guarantee. In the *local* setting, by contrast, no *bounded-factor* approximation exists — already for a *single* sample point — which marks a sharp global/local gap.

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

We additionally establish a matching *lower bound* on the approximation factor of Theorem 3, for the case of an empirical distribution without any independence assumption (see Appendix M).

Cardinally minimal sufficient reasons. In contrast to the submodular set cover problem — which governs cardinally minimal global *contrastive* reasons and admits strong approximations — the supermodular variant governing global *sufficient* reasons is harder to approximate, yet still admits guarantees once the objective has bounded curvature (Shi et al., 2021). The *total curvature* of $v : 2^n \rightarrow \mathbb{R}$ is $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$. Invoking the guarantees of Shi et al. (2021), we show that Algorithm 2 attains an $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation.

Under a fixed empirical distribution this factor is again constant. Relative to the tighter $\mathcal{O}(\ln(|D|))$ bound available for contrastive reasons, the sufficient case incurs an additional $\frac{1}{1-k^f}$ term, but still yields a constant-factor approximation. The local variant, by contrast, remains inapproximable: no *bounded-factor* approximation exists even when $|D| = 1$.

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{uff}}^g([n])}{\min_{i \in [n]} v_{\text{uff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Taken together, these results reinforce Subsection 7.1, which established the tractability of computing subset-minimal global explanations against the intractability of their local counterparts. We have now shown, further, that approximating cardinally minimal global explanations is likewise tractable, whereas the corresponding local problems are inapproximable.

8 LIMITATIONS AND FUTURE WORK

While our principal results — in particular those concerning intrinsic properties of the value functions — are model-agnostic, we also instantiate them to obtain concrete complexity results under specific model classes (e.g., neural networks), distributional assumptions (e.g., empirical distributions), and explanation definitions admitted by our framework. Other settings naturally remain open. Nonetheless, we contend that our results yield substantive insight into the foundational structure of explanations, together with new tractability and intractability results that constitute a basis for analyzing a broader class of explainability settings in future work.

More specifically, our analysis raises several concrete open theoretical questions:

1. First, whether the tractability results we obtain for global explainability can be tightened for structurally simpler model classes — decision trees, linear models, or other classes with inherently tractable structure. Although local explanations for such models are typically intractable (Arenas et al., 2022; Subercaseaux et al., 2025), the global explanation forms studied here possess strong structural properties that yield markedly improved complexity guarantees, and these may admit further improvement. Tightening the approximation factors under empirical distributions is a further open direction.
2. Second, when the required expectations cannot be evaluated *exactly*, whether alternative estimators — Fully Polynomial Randomized Approximation Schemes (FPRAS) for classes such as DNF formulas (Meel et al., 2019) (hence for tree ensembles), or Monte Carlo estimators of the kind used in (Subercaseaux et al., 2025) — suffice to approximate them. The central technical challenge is to characterize the approximation-error tolerance under which the minimality guarantees provably persist.
3. Finally, whether alternative feature-importance criteria — for instance, loss measures such as the KL divergence (Conmy et al., 2023) — exhibit the same monotonicity, submodularity, or supermodularity structure in the local-to-global transition. Establishing such structure would extend efficient feature selection to these alternative criteria.

9 CONCLUSION

We have introduced a unified framework for analyzing diverse explanation forms and used it to expose a sharp dichotomy between local and global sufficient and contrastive reasons. Whereas the *local* variants possess none of the structural properties central to combinatorial optimization — monotonicity, submodularity, or supermodularity — we prove that their *global* counterparts possess exactly these properties: (i) *monotonicity*, (ii) *submodularity* for contrastive reasons, and (iii) *supermodularity* for sufficient reasons. These structural results underpin a series of complexity separations establishing that global explanations with provable guarantees are efficiently computable even for expressive model classes such as neural networks, while the corresponding local explanations remain NP-hard even under substantially simplified assumptions. In sum, our results characterize foundational structural properties of explanations and delineate the tractable and intractable frontiers of their computation, opening new directions for future research.

REFERENCES

- Federico Adolfi, Martina G Vilas, and Todd Wareham. Complexity-Theoretic Limits on the Promises of Artificial Neural Network Reverse-Engineering. In *Proc. 46th Annual Meeting of the Cognitive Science Society (CogSci)*, 2024.
- Federico Adolfi, Martina Vilas, and Todd Wareham. The Computational Complexity of Circuit Discovery for Inner Interpretability. In *Proc. 13th Int. Conf. on Learning Representations (ICLR)*, 2025.

540 Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, Reza Shahbazian, and
541 Irina Trubitsyna. Even-If Explanations: Formal Foundations, Priorities and Complexity. In *Proc.*
542 *39th AAAI Conf. on Artificial Intelligence*, pp. 15347–15355, 2025.

543

544 Ibrahim M Almanjahie, Zouaoui Chikr Elmezouar, Ali Laksaci, and Mustapha Rachdi. KNN Local
545 Linear Estimation of the Conditional Cumulative Distribution Function: Dependent Functional
546 Data Case. *Comptes Rendus. Mathématique*, 356(10):1036–1039, 2018.

547 David Alvarez Melis and Tommi Jaakkola. Towards Robust Interpretability with Self-Explaining
548 Neural Networks. In *Proc. 32nd Conf. on Neural Information Processing Systems (NeurIPS)*, 2018.

549

550 Leila Amgoud and Martin Cooper. Axiomatic Foundations of Counterfactual Explanations. 2026.
551 Technical Report. <https://arxiv.org/abs/2602.04028>.

552

553 Guy Amir, Shahaf Bassan, and Guy Katz. Hard to Explain: On the Computational Hardness of
554 In-Distribution Model Interpretation. In *Proc. 27th European Conf. on Artificial Intelligence (ECAI)*,
555 2024.

556 Alaa Anani, Tobias Lorenz, Mario Fritz, and Bernt Schiele. Pixel-level Certified Explanations via
557 Randomized Smoothing. In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025.

558

559 Marcelo Arenas, D. Baez, P. Barceló, J. Pérez, and B. Subercaseaux. Foundations of Symbolic
560 Languages for Model Interpretability. In *Proc. 34th Int. Conf. on Advances in Neural Information*
561 *Processing Systems (NeurIPS)*, pp. 11690–11701, 2021.

562

563 Marcelo Arenas, P. Barceló, M. Romero, and B. Subercaseaux. On Computing Probabilistic Explana-
564 tions for Decision Trees. In *Proc. 35th Int. Conf. on Advances in Neural Information Processing*
565 *Systems (NeurIPS)*, pp. 28695–28707, 2022.

566

567 Marcelo Arenas, Pablo Barceló, Leopoldo Bertossi, and Mikaël Monet. On the Complexity of SHAP-
568 Score-Based Explanations: Tractability via Knowledge Compilation and Non-Approximability
569 Results. *Journal of Machine Learning Research (JMLR)*, 24(63):1–58, 2023.

570

571 Gilles Audemard, S. Bellart, L. Bounia, F. Koriche, J. Lagniez, and P. Marquis. On the Computa-
572 tional Intelligibility of Boolean Classifiers. In *Proc. 18th Int. Conf. on Principles of Knowledge*
573 *Representation and Reasoning (KR)*, pp. 74–86, 2021.

574

575 Gilles Audemard, Steve Bellart, Louenas Bounia, Frédéric Koriche, Jean-Marie Lagniez, and Pierre
576 Marquis. On Preferred Abductive Explanations for Decision Trees and Random Forests. In *Proc.*
577 *31st Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 643–650, 2022a.

578

579 Gilles Audemard, Steve Bellart, Louenas Bounia, Frédéric Koriche, Jean-Marie Lagniez, and Pierre
580 Marquis. Trading Complexity for Sparsity in Random Forest Explanations. In *Proc. 36th AAAI*
581 *Conf. on Artificial Intelligence*, number 5, pp. 5461–5469, 2022b.

582

583 Gilles Audemard, Jean-Marie Lagniez, Pierre Marquis, and Nicolas Szczepanski. Computing
584 Abductive Explanations for Boosted Trees. In *Proc. 26th Int. Conf. on Artificial Intelligence and*
585 *Statistics (AISTATS)*, pp. 4699–4711, 2023.

586

587 Steve Azzolin, Sagar Malhorta, Andrea Passerini, and Stefano Teso. Beyond Topological Self-
588 Explainable GNNs: A Formal Explainability Perspective. In *Proc. 42nd Int. Conf. on Machine*
589 *Learning (ICML)*, 2025.

590

591 Esma Balkir, Isar Nejadgholi, Kathleen C Fraser, and Svetlana Kiritchenko. Necessity and Sufficiency
592 for Explaining Text Classifiers: A Case Study in Hate Speech Detection. In *Proc. Conf. of the*
593 *North American Chapter of the Association for Computational Linguistics: Human Language*
Technologies (NAACL-HLT), pp. 2672–2686, 2022.

Pablo Barceló, M. Monet, J. Pérez, and B. Subercaseaux. Model Interpretability through the Lens of
Computational Complexity. In *Proc. 33rd Int. Conf. on Advances in Neural Information Processing*
Systems (NeurIPS), pp. 15487–15498, 2020.

594 Pablo Barceló, Alexander Kozachinskiy, Miguel Romero, Bernardo Subsercaseaux, and José Verschae.
595 Explaining k-Nearest Neighbors: Abductive and Counterfactual Explanations. In *Proc. 3rd ACM*
596 *on Management of Data*, pp. 1–26, 2025.

597

598 Shahaf Bassan and G. Katz. Towards Formal XAI: Formally Approximate Minimal Explanations
599 of Neural Networks. In *Proc. 29th Int. Conf. on Tools and Algorithms for the Construction and*
600 *Analysis of Systems (TACAS)*, pp. 187–207, 2023.

601

602 Shahaf Bassan, Guy Amir, Davide Corsi, Idan Refaeli, and Guy Katz. Formally Explaining Neural
603 Networks Within Reactive Systems. In *Proc. 23rd Int. Conf. on Formal Methods in Computer-Aided*
604 *Design (FMCAD)*, pp. 1–13, 2023.

605

606 Shahaf Bassan, Guy Amir, and Guy Katz. Local vs. Global Interpretability: A Computational
607 Complexity Perspective. In *Proc. 41st Int. Conf. on Machine Learning (ICML)*, 2024.

608

609 Shahaf Bassan, Guy Amir, Meirav Zehavi, and Guy Katz. What makes an Ensemble (Un) Inter-
610 pretable? In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025a.

611

612 Shahaf Bassan, Yizhak Yisrael Elboher, Tobias Ladner, Matthias Althoff, and Guy Katz. Explaining,
613 Fast and Slow: Abstraction and Refinement of Provable Explanations. In *Proc. 42nd Int. Conf. on*
614 *Machine Learning (ICML)*, 2025b.

615

616 Shahaf Bassan, Ron Eliav, and Shlomit Gur. Explain Yourself, Briefly! Self-Explaining Neural
617 Networks with Concise Sufficient Reasons. In *Proc. 13th Int. Conf. on Learning Representations*
618 *(ICLR)*, 2025c.

619

620 Shahaf Bassan, Shlomit Gur, Sergey Zeltyn, Konstantinos Mavrogiorgos, Ron Eliav, and Dimosthenis
621 Kyriazis. Self-Explaining Neural Networks for Business Process Monitoring. 2025d. Technical
622 Report. <https://arxiv.org/abs/2503.18067>.

623

624 Shahaf Bassan, Michal Moshkovitz, and Guy Katz. Additive Models Explained: A Computational
625 Complexity Approach. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*,
626 2025e.

627

628 Shahaf Bassan, Yizhak Yisrael Elboher, Tobias Ladner, Volkan Şahin, Jan Kretinsky, Matthias Althoff,
629 and Guy Katz. Provably Explaining Neural Additive Models. In *Proc. 14th Int. Conf. on Learning*
630 *Representations (ICLR)*, 2026.

631

632 Robi Bhattacharjee and Ulrike Luxburg. Auditing Local Explanations is Hard. In *Proc. 38th Conf.*
633 *on Neural Information Processing Systems (NeurIPS)*, pp. 18593–18632, 2024.

634

635 Blair Bilodeau, Natasha Jaques, Pang Wei Koh, and Been Kim. Impossibility Theorems for Feature
636 Attribution. *Proc. of the National Academy of Sciences (PNAS)*, 121(2):e2304406120, 2024.

637

638 Guy Blanc, Jane Lange, and Li-Yang Tan. Provably Efficient, Succinct, and Precise Explanations.
639 In *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp.
640 6129–6141, 2021.

641

642 Guy Blanc, Caleb Koch, Jane Lange, and Li-Yang Tan. A Query-Optimal Algorithm for Finding
643 Counterfactuals. In *Proc. 39th Int. Conf. on Machine Learning (ICML)*, pp. 2075–2090, 2022.

644

645 Ryma Boumazouza, Fahima Cheikh-Alili, Bertrand Mazure, and Karim Tabia. ASTERYX: A model-
646 Agnostic SaT-basEd appRoach for SYmbolic and Score-Based EXplanations. In *Proc. 30th ACM*
647 *Int. Conf. on Information & Knowledge Management (CIKM)*, pp. 120–129, 2021.

648

649 Ryma Boumazouza, Raya Elsaleh, Melanie Ducoffe, Shahaf Bassan, and Guy Katz. FAME: For-
650 mal Abstract Minimal Explanation for Neural Networks. In *Proc. 14th Int. Conf. on Learning*
651 *Representations (ICLR)*, 2026.

652

653 Louenas Bounia. Enhancing the Intelligibility of Boolean Decision Trees with Concise and Reliable
654 Probabilistic Explanations. In *Proc. 20th Int. Conf. on Information Processing and Management of*
655 *Uncertainty in Knowledge-Based Systems (IPMU)*, pp. 205–218, 2024.

648 Louenas Bounia. Using Submodular Optimization to Approximate Minimum-Size Abductive Path
649 Explanations for Tree-Based Models. In *Proc. 41st Conf. on Uncertainty in Artificial Intelligence*
650 *(UAI)*, 2025.

651 Louenas Bounia and Frederic Koriche. Approximating Probabilistic Explanations via Supermodular
652 Minimization. In *Proc. 39th Conf. on Uncertainty in Artificial Intelligence (UAI)*, pp. 216–225,
653 2023.

654 Marco Calautti, Enrico Malizia, and Cristian Molinaro. On the Complexity of Global Necessary
655 Reasons to Explain Classification. 2025. Technical Report. [https://arXiv.org/abs/](https://arXiv.org/abs/2501.06766)
656 2501.06766.

657 Emilio Carrizosa, Jasone Ramírez-Ayerbe, and Dolores Romero Morales. Mathematical Optimization
658 Modelling for Group Counterfactual Explanations. *European Journal of Operational Research*
659 *(EJOR)*, 319(2):399–412, 2024.

660 Brandon Carter, Jonas Mueller, Siddhartha Jain, and David Gifford. What Made You Do This?
661 Understanding Black-Box Decisions with Sufficient Input Subsets. In *Proc. 22nd Int. Conf. on*
662 *Artificial Intelligence and Statistics (AISTATS)*, pp. 567–576, 2019.

663 Brandon Carter, Siddhartha Jain, Jonas W Mueller, and David Gifford. Overinterpretation Reveals
664 Image Classification Model Pathologies. In *Proc. 35th Conf. on Neural Information Processing*
665 *Systems (NeurIPS)*, pp. 15395–15407, 2021.

666 Wenjing Chen and Victoria Crawford. Bicriteria Approximation Algorithms for the Submodular
667 Cover Problem. In *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp.
668 72705–72716, 2023.

669 Arthur Conmy, Augustine Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-
670 Alonso. Towards Automated Circuit Discovery for Mechanistic Interpretability. *Proc. 37th Conf.*
671 *on Neural Information Processing Systems (NeurIPS)*, pp. 16318–16352, 2023.

672 Yves Crama and Peter L Hammer. *Boolean Functions: Theory, Algorithms, and Applications*.
673 Cambridge University Press, 2011.

674 Adnan Darwiche. Logic for Explainable AI. In *Proc. 38th ACM/IEEE Symposium on Logic in*
675 *Computer Science (LICS)*, pp. 1–11, 2023.

676 Adnan Darwiche and Auguste Hirth. On the Reasons Behind Decisions. In *Proc. 24th European*
677 *Conf. on Artificial Intelligence (ECAI)*, pp. 712–720, 2020.

678 Adnan Darwiche and Chunxi Ji. On the Computation of Necessary and Sufficient Explanations. In
679 *Proc. 36th AAAI Conf. on Artificial Intelligence*, pp. 5582–5591, 2022.

680 Sanjoy Dasgupta, Nave Frost, and Michal Moshkovitz. Framework for Evaluating Faithfulness of
681 Local Explanations. In *Proc. 39th Int. Conf. on Machine Learning (ICML)*, pp. 4794–4815, 2022.

682 Ian Davidson, Nicolás Kennedy, and SS Ravi. CXAD: Contrastive Explanations for Anomaly
683 Detection: Algorithms, Complexity Results and Experiments. *Transactions on Machine Learning*
684 *Research (TMLR)*, 2025.

685 Alessandro De Palma, Greta Dolcetti, and Caterina Urban. Faster Verified Explanations for Neural
686 Networks. 2025. Technical Report. <https://arXiv.org/abs/2512.00164>.

687 Amit Dhurandhar, Pin-Yu Chen, Ronny Luss, Chun-Chen Tu, Paishun Ting, Karthikeyan Shanmugam,
688 and Payel Das. Explanations Based on the Missing: Towards Contrastive Explanations with
689 Pertinent Negatives. *Proc. 32nd Conf. on Neural Information Processing Systems (NeurIPS)*, 2018.

690 Hidde Fokkema, Rianne De Heide, and Tim Van Erven. Attribution-Based Explanations that Provide
691 Recourse Cannot be Robust. *Journal of Machine Learning Research (JMLR)*, 24(360):1–37, 2023.

692 Hidde Fokkema, Damien Garreau, and Tim van Erven. The Risks of Recourse in Binary Classification.
693 In *Proc. 27th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, pp. 550–558, 2024.

-
- Tobias Geibinger, Reijo Jaakkola, Antti Kuusisto, Xinghan Liu, and Miikka Vilander. Why This and Not That? A Logic-based Framework for Contrastive Explanations. In *Proc. 19th European Conf. on Logics in Artificial Intelligence (JELIA)*, pp. 45–60. Springer, 2025.
- Joshua Gröber and Ostap Okhrin. Copulae: An Overview and Recent Developments. *Wiley Interdisciplinary Reviews: Computational Statistics*, 14(3):e1557, 2022.
- Riccardo Guidotti. Counterfactual Explanations and How to Find Them: Literature Review and Benchmarking. *Data Mining and Knowledge Discovery (DMKD)*, 38(5):2770–2824, 2024.
- Hyukjun Gweon, Matthias Schonlau, and Stefan H Steiner. The K Conditional Nearest Neighbor Algorithm for Classification and Class Probability Estimation. *PeerJ Computer Science*, 5:e194, 2019.
- Itamar Hadad, Shahaf Bassan, and Guy Katz. Formal Mechanistic Interpretability: Automated Circuit Discovery with Provable Guarantees. In *Proc. 14th Int. Conf. on Learning Representations (ICLR)*, 2026.
- Ouns El Harzli, Bernardo Cuenca Grau, and Ian Horrocks. Cardinality-Minimal Explanations for Monotonic Neural Networks. In *Proc. 32nd Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 3677–3685, 2023.
- W Keith Hastings. Monte Carlo Sampling Methods using Markov Chains and their Applications. *Biometrika*, 57:97–109, 1970.
- Rongjie Huang, Max W. Y. Lam, Jun Wang, Dan Su, Dong Yu, Yi Ren, and Zhou Zhao. FastDiff: A Fast Conditional Diffusion Model for High-Quality Speech Synthesis. In *Proc. 31st Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 4157–4163, 2022.
- Xuanxiang Huang, Yacine Izza, Alexey Ignatiev, and Joao Marques-Silva. On Efficiently Explaining Graph-Based Classifiers. In *Proc. 18th Int. Conf. on Principles of Knowledge Representation and Reasoning (KR)*, 2021.
- Alexey Ignatiev, Nina Narodytska, and Joao Marques-Silva. Abduction-Based Explanations for Machine Learning Models. In *Proc. 33rd AAAI Conference on Artificial Intelligence*, pp. 1511–1519, 2019.
- Alexey Ignatiev, Martin Cooper, Mohamed Siala, Emmanuel Hebrard, and Joao Marques-Silva. Towards Formal Fairness in Machine Learning. In *Proc. 26th Int. Conf. on Principles and Practice of Constraint Programming (CP)*, pp. 846–867, 2020a.
- Alexey Ignatiev, Nina Narodytska, Nicholas Asher, and Joao Marques-Silva. From Contrastive to Abductive Explanations and Back Again. In *Proc. Int. Conf. of the Italian Association for Artificial Intelligence (AIxAI)*, pp. 335–355, 2020b.
- Alexey Ignatiev, Yacine Izza, Peter J Stuckey, and Joao Marques-Silva. Using MaxSAT for Efficient Explanations of Tree Ensembles. In *Proc. 36th AAAI Conference on Artificial Intelligence*, pp. 3776–3785, 2022.
- Rishabh K Iyer and Jeff A Bilmes. Submodular Optimization with Submodular Cover and Submodular Knapsack Constraints. *Proc. 27th Conf. on Neural Information Processing Systems (NeurIPS)*, 2013.
- Yacine Izza and Joao Marques-Silva. On Explaining Random Forests with SAT. In *Proc. 30th Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2021.
- Yacine Izza, Xuanxiang Huang, Alexey Ignatiev, Nina Narodytska, Martin Cooper, and Joao Marques-Silva. On Computing Probabilistic Abductive Explanations. *Int. Journal of Approximate Reasoning (IJAR)*, 159:108939, 2023.
- Yacine Izza, Xuanxiang Huang, Antonio Morgado, Jordi Planes, Alexey Ignatiev, and Joao Marques-Silva. Distance-Restricted Explanations: Theoretical Underpinnings & Efficient implementation. In *Proc. 21st Int. Conf. on Principles of Knowledge Representation and Reasoning (KR)*, pp. 475–486, 2024a.

756 Yacine Izza, Kuldeep S Meel, and Joao Marques-Silva. Locally-Minimal Probabilistic Explanations.
757 In *Proc. 27th European Conf. on Artificial Intelligence (ECAI)*, 2024b.

758

759 Reijo Jaakkola, Tomi Janhunen, Antti Kuusisto, Masood Feyzbakhsh Rankooh, and Miikka Vilander.
760 Explainability via Short Formulas: the Case of Propositional Logic with Implementation. *Journal*
761 *of Artificial Intelligence Research (JAIR)*, 83, 2025.

762 Alon Jacovi, Swabha Swayamdipta, Shauli Ravfogel, Yanai Elazar, Yejin Choi, and Yoav Goldberg.
763 Contrastive Explanations for Model Interpretability. In *Proc. Conf. Conference on Empirical*
764 *Methods in Natural Language Processing (EMNLP)*, pp. 1597–1611, 2021.

765

766 Helen Jin, Anton Xue, Weiqiu You, Surbhi Goel, and Eric Wong. Probabilistic Stability Guarantees
767 for Feature Attributions. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*,
768 2025.

769 Amir-Hossein Karimi, Bernhard Schölkopf, and Isabel Valera. Algorithmic recourse: from counter-
770 factual explanations to interventions. In *Proc. 4th ACM Conf. on Fairness, Accountability, and*
771 *Transparency (FAccT)*, pp. 353–362, 2021.

772

773 Eoin M Kenny and Mark T Keane. On Generating Plausible Counterfactual and Semi-Factual
774 Explanations for Deep Learning. In *Proc. 35th AAAI Conf. on Artificial Intelligence*, number 13,
775 pp. 11575–11585, 2021.

776 Alexander Kozachinskiy. Inapproximability of Sufficient Reasons for Decision Trees. 2023. Technical
777 Report. <https://arXiv.org/abs/2304.02781>.

778

779 I Elizabeth Kumar, Suresh Venkatasubramanian, Carlos Scheidegger, and Sorelle Friedler. Problems
780 with Shapley-Value-Based Explanations as Feature Importance Measures. In *Proc. 37th Int. Conf.*
781 *on Machine Learning (ICML)*, pp. 5491–5500, 2020.

782 Emanuele La Malfa, Agnieszka Zbrzezny, Rhiannon Michelmor, Nicola Paoletti, and Marta
783 Kwiakowska. On Guaranteed Optimal Robust Explanations for NLP Models. In *Proc. Int.*
784 *Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 2658–2665, 2021.

785

786 Eduardo Sany Laber. The Computational Complexity of some Explainable Clustering Problems.
787 *Information Processing Letters*, 184:106437, 2024.

788

789 Shuai Li, Yingjie Zhang, Hongtu Zhu, Christina Wang, Hai Shu, Ziqi Chen, Zhuoran Sun, and
790 Yanfeng Yang. K-Nearest-Neighbor Local Sampling Based Conditional Independence Testing.
791 *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 23321–23344, 2023.

792 Gianluigi Lopardo, Frédéric Precioso, and Damien Garreau. A Sea of Words: An In-Depth Analysis
793 of Anchors for Text Data. In *Proc. 26th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*
794 2023, 2023.

795

796 Scott M Lundberg and Su-In Lee. A Unified Approach to Interpreting Model Predictions. *Proc. 31st*
797 *Conf. on Neural Information Processing Systems (NeurIPS)*, 2017.

798 Joao Marques-Silva and Alexey Ignatiev. Delivering Trustworthy AI through formal XAI. In *Proc.*
799 *36th AAAI Conf. on Artificial Intelligence*, pp. 12342–12350, 2022.

800

801 Joao Marques-Silva and Alexey Ignatiev. No Silver Bullet: Interpretable ML Models Must be
802 Explained. *Frontiers in Artificial Intelligence*, 6:1128212, 2023.

803 Joao Marques-Silva, Thomas Gerspacher, Martin Cooper, Alexey Ignatiev, and Nina Narodytska.
804 Explaining Naive Bayes and Other Linear Classifiers with Polynomial Time and Delay. In *Proc.*
805 *33rd Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 20590–
806 20600, 2020.

807

808 Joao Marques-Silva, Thomas Gerspacher, Martin Cooper, Alexey Ignatiev, and Nina Narodytska.
809 Explanations for Monotonic Classifiers. In *Proc. Int. Conf. on Machine Learning (ICML)*, pp.
7469–7479, 2021.

-
- Reda Marzouk and Colin De La Higuera. On the Tractability of SHAP Explanations under Markovian Distributions. In *Proc. 41st Int. Conf. on Machine Learning (ICML)*, pp. 34961–34986, 2024.
- Reda Marzouk, Shahaf Bassan, and Guy Katz. SHAP Meets Tensor Networks: Provably Tractable Explanations with Parallelism. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*, 2025a.
- Reda Marzouk, Shahaf Bassan, Guy Katz, and De la Higuera. On the Computational Tractability of the (Many) Shapley Values. In *Proc. 28th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, 2025b.
- Kuldeep S Meel, Aditya A Shrotri, and Moshe Y Vardi. Not all FPRASs are Equal: Demystifying FPRASs for DNF-Counting. *Constraints*, 24(3):211–233, 2019.
- Ramaravind K Mothilal, Amit Sharma, and Chenhao Tan. Explaining Machine Learning Classifiers Through Diverse Counterfactual Explanations. In *Proc. 3rd Conf. on Fairness, Accountability, and Transparency (FAccT)*, pp. 607–617, 2020.
- Vidya Muthukumar, Tejaswini Pedapati, Nalini Ratha, Prasanna Sattigeri, Chai-Wah Wu, Brian Kingsbury, Abhishek Kumar, Samuel Thomas, Aleksandra Mojsilovic, and Kush R Varshney. Understanding Unequal Gender Classification Accuracy from Face Images. 2018. Technical Report. <https://arXiv.org/abs/1812.00099>.
- Sebastian Ordyniak, Giacomo Paesani, and Stefan Szeider. The Parameterized Complexity of Finding Concise Local Explanations. In *Proc. 32nd Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2023.
- Sebastian Ordyniak, Giacomo Paesani, Mateusz Rychlicki, and Stefan Szeider. Explaining Decisions in ML Models: A Parameterized Complexity Analysis. In *Proc. 21st Int. Conf. on Principles of Knowledge Representation and Reasoning (KR)*, pp. 563–573, 2024.
- Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. Anchors: High-Precision Model-Agnostic Explanations. In *Proc. 32nd AAAI Conf. on Artificial Intelligence*, 2018.
- Majun Shi, Zishen Yang, and Wei Wang. Minimum Non-Submodular Cover Problem with Applications. *Applied Mathematics and Computation*, 410:126442, 2021.
- Andy Shih, Arthur Choi, and Adnan Darwiche. A Symbolic Approach to Explaining Bayesian Network Classifiers. In *Proc. 27th Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 5103–5111, 2018.
- Bernardo Subercaseaux, Marcelo Arenas, and Kuldeep S Meel. Probabilistic Explanations for Linear Models. In *Proc. 39th AAAI Conf. on Artificial Intelligence*, pp. 20655–20662, 2025.
- Guy Van den Broeck, Anton Lykov, Maximilian Schleich, and Dan Suciu. On the Tractability of SHAP Explanations. *Journal of Artificial Intelligence Research (JAIR)*, 74:851–886, 2022.
- Gal Vardi, Gilad Yehudai, and Ohad Shamir. On the Optimal Memorization Power of ReLU Neural Networks. In *Proc. 10th Int. Conf. on Learning Representations, (ICLR)*, 2022.
- Vijay V Vazirani. *Approximation Algorithms*. Springer, 2001.
- Sahil Verma, Varich Boonsanong, Minh Hoang, Keegan Hines, John Dickerson, and Chirag Shah. Counterfactual Explanations and Algorithmic Recourses for Machine Learning: A Review. *ACM Computing Surveys*, 56(12):1–42, 2024.
- Stephan Wäldchen, Jan Macdonald, Sascha Hauch, and Gitta Kutyniok. The Computational Complexity of Understanding Binary Classifier Decisions. *Journal of Artificial Intelligence Research (JAIR)*, 70:351–387, 2021.
- E. Wang, P. Khosravi, and G. Van den Broeck. Probabilistic Sufficient Explanations. In *Proc. 30th Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2021a.

-
- 864 S. Wang, H. Zhang, K. Xu, X. Lin, S. Jana, C. Hsieh, and Z. Kolter. Beta-Crown: Efficient Bound
865 Propagation with Per-Neuron Split Constraints for Neural Network Robustness Verification. In
866 *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp.
867 29909–29921, 2021b.
- 868 Greta Warren, Eoin Delaney, Christophe Guéret, and Mark T Keane. Explaining Multiple Instances
869 Counterfactually: User Tests of Group-Counterfactuals for XAI. In *Proc. 32nd Int. Conf. on*
870 *Case-Based Reasoning (ICCBR)*, pp. 206–222, 2024.
- 871 Ingo Wegener. The Complexity of Symmetric Boolean Functions. *Computation Theory and Logic*,
872 pp. 433–442, 2005.
- 873 Laurence A Wolsey. An Analysis of the Greedy Algorithm for the Submodular Set Covering Problem.
874 *Combinatorica*, 2(4):385–393, 1982.
- 875 Haoze Wu, Omri Isac, Aleksandar Zeljić, Teruhiro Tagomori, Matthew Daggitt, Wen Kokke, Idan
876 Refaeli, Guy Amir, Kyle Julian, Shahaf Bassan, et al. Marabou 2.0: A Versatile Formal Analyzer
877 of Neural Networks. In *Proc. 36th Int. Conf. on Computer Aided Verification (CAV)*, pp. 249–264,
878 2024a.
- 879 Min Wu, Haoze Wu, and Clark Barrett. Verix: Towards Verified Explainability of Deep Neural
880 Networks. In *Proc. 36th Int. Conf. on Advances in Neural Information Processing Systems*
881 *(NeurIPS)*, 2024b.
- 882 Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling Tabular
883 Data Using Conditional GAN. *Proc. 33rd Conf. on Neural Information Processing Systems*
884 *(NeurIPS)*, 2019.
- 885 Anton Xue, Rajeev Alur, and Eric Wong. Stability Guarantees for Feature Attributions with Multi-
886 plicative Smoothing. *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp.
887 62388–62413, 2023.
- 888 Kayo Yin and Graham Neubig. Interpreting Language Models with Contrastive Explanations. In
889 *Proc. Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 184–198, 2022.
- 890 Jinqiang Yu, Alexey Ignatiev, Peter Stuckey, Nina Narodytska, and Joao Marques-Silva. Eliminating
891 the Impossible, Whatever Remains Must Be True: On Extracting and Applying Background
892 Knowledge in the Context of Formal Explanations. In *Proc. 37th AAAI Conf. on Artificial*
893 *Intelligence*, pp. 4123–4131, 2023.
- 894
895
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897
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Appendix

The appendix contains formalizations and proofs that were mentioned throughout the paper:

Appendix A contains extended related work on Formal XAI.

Appendix B contains the formalizations of the models and distributions used in this work.

Appendix C contains the proof of Proposition 1.

Appendix D contains the proof of Proposition 2.

Appendix E contains the proof of Proposition 3.

Appendix F contains the proof of Proposition 4.

Appendix G contains the proof of Proposition 5.

Appendix H contains the proof of Theorem 1.

Appendix I contains the proof of Proposition 6.

Appendix J contains the proof of Theorem 2.

Appendix K contains the proof of Theorem 3.

Appendix L contains the proof of Theorem 4.

Appendix O contains an LLM usage disclosure.

A EXTENDED RELATED WORK ON FORMAL XAI

Our work belongs to the line of research on *formal explainable AI* (formal XAI) (Marques-Silva & Ignatiev, 2022), which studies explanations endowed with provable guarantees (Yu et al., 2023; Darwiche & Ji, 2022; Darwiche, 2023; Shih et al., 2018; Azzolin et al., 2025; Audemard et al., 2021; Calautti et al., 2025). A central theme in this area is the computational complexity of generating such guaranteed explanations (Barceló et al., 2020; ?; Marzouk et al., 2025a;b; Marzouk & De La Higuera, 2024; Bassan et al., 2025e; Amir et al., 2024; Blanc et al., 2021; 2022; Jaakkola et al., 2025; Amgoud & Cooper, 2026; Geibinger et al., 2025). Because computing formal explanations is typically intractable for expressive model classes such as neural networks (Adolfi et al., 2024; Barceló et al., 2020) and tree ensembles (Ordyniak et al., 2024; Bassan et al., 2025a), much of this literature restricts attention to more tractable classes (Marques-Silva & Ignatiev, 2023), including decision trees (Bounia, 2025; Arenas et al., 2022; Bounia, 2024; Bounia & Koriche, 2023), monotonic models (Marques-Silva et al., 2021; Harzli et al., 2023), linear models (Marques-Silva et al., 2020; Subercaseaux et al., 2025), and additive models (Bassan et al., 2026).

Beyond complexity-theoretic analysis, a substantial body of work computes formal explanations in practice via automated reasoning tools — MaxSAT and MILP solvers for tree ensembles (Audemard et al., 2023; 2022a; Ignatiev et al., 2022; Boumazouza et al., 2021), and neural network verification frameworks (Wang et al., 2021b; Wu et al., 2024a) for deriving certified explanations of neural networks (Wu et al., 2024b; Bassan & Katz, 2023; Bassan et al., 2023; Izza et al., 2024a; Hadad et al., 2026; Ignatiev et al., 2019). A complementary line of work seeks to mitigate the intrinsic computational cost through abstractions (De Palma et al., 2025; Bassan et al., 2025b; Boumazouza et al., 2026), relaxations such as smoothing (Xue et al., 2023; Anani et al., 2025; Jin et al., 2025), and training-time interventions (Alvarez Melis & Jaakkola, 2018; Bassan et al., 2025c;d).

Finally, a related strand of formal XAI applies combinatorial and submodular optimization techniques to the computation of *local* probabilistic sufficient reasons (Bounia, 2025; Bounia & Koriche, 2023), chiefly for decision tree classifiers. As observed both in that literature and here, the value-function objective underlying these local variants is not submodular. Greedy algorithms motivated by submodular optimization nonetheless serve as effective heuristics and empirically yield high-quality explanations, but without formal guarantees. Our work, by contrast, establishes a rigorous correspondence between combinatorial/submodular optimization objectives and the *global* variants of sufficient and contrastive reasons — a correspondence that furnishes provable guarantees and admits a broad range of principled algorithmic instantiations.

B MODEL AND DISTRIBUTION FORMALIZATIONS

This section provides the formal definitions of the model classes and distribution classes referenced throughout the paper. Subsection B.1 formalizes the model families, and Subsection B.2 formalizes the distribution classes.

B.1 MODEL FORMALIZATIONS

This subsection formalizes the three base model classes analyzed throughout the paper: (i) (axis-aligned) decision trees, (ii) linear classifiers, and (iii) neural networks with ReLU activations.

Decision Trees. We define a decision tree (DT) as a directed acyclic graph encoding a function $f : \mathbb{F} \rightarrow [c]$, where $c \in \mathbb{N}$ is the number of classes. The encoding is specified by the following conditions: (i) each internal node v is labeled with a distinct binary input feature from $\{1, \dots, n\}$; (ii) each internal node v has at most k outgoing edges, each labeled with a distinct value in $[k]$ assigned to v ; (iii) along any root-to-leaf path α , each variable is tested at most once; (iv) each leaf is labeled with a class in $[c]$. Consequently, any input assignment $\mathbf{x} \in \mathbb{F}$ determines a unique root-to-leaf path α , and the output $f(\mathbf{x})$ is the class label $i \in [c]$ of the reached leaf. The size $|f|$ of the DT is the total number of edges in the graph. To permit flexible modeling, the variable ordering may differ across distinct paths α and α' , subject only to the no-repetition constraint along each individual path.

Neural Networks. We state our *hardness* results for neural networks with ReLU activations, whereas our *tractability* results (i.e., polynomial-time algorithms) hold for any architecture admitting polynomial-time inference — a standard assumption. Consequently, every tractable/intractable separation we establish transfers to *any* neural architecture at least as expressive as a standard feed-forward ReLU network, including widely used models such as ResNets, CNNs, Transformers, and diffusion models. Moreover, any ReLU network is expressible as a fully connected network by assigning zero weights and biases to absent connections; following standard conventions (Barceló et al., 2020; Bassan et al., 2024; Adolphi et al., 2025), we therefore assume full connectivity, i.e., our analysis concerns multi-layer perceptrons (MLPs). Formally, an MLP f comprises $t - 1$ *hidden layers* $g^{(j)}$ for $j = 1, \dots, t - 1$ together with an output layer $g^{(t)}$, each defined by the recursion:

$$g^{(j)} := \sigma^{(j)}(g^{(j-1)}W^{(j)} + b^{(j)}) \quad (9)$$

where $W^{(j)}$, $b^{(j)}$, and $\sigma^{(j)}$ denote, respectively, the weight matrix, bias vector, and activation function of the j -th layer. The model is thus parameterized by t weight matrices $(W^{(1)}, \dots, W^{(t)})$, t bias vectors $(b^{(1)}, \dots, b^{(t)})$, and t activation functions $(\sigma^{(1)}, \dots, \sigma^{(t)})$.

The input layer is $g^{(0)} = \mathbf{x} \in \{0, 1\}^n$, the binary input vector. The layer widths are given by a sequence of positive integers $d_0, \dots, d_t$, and the (learned) parameters satisfy $W^{(j)} \in \mathbb{Q}^{d_{j-1} \times d_j}$ and $b^{(j)} \in \mathbb{Q}^{d_j}$. Since f acts as a binary classifier over n features, we set $d_0 = n$ and $d_t = 1$. The hidden layers use the ReLU activation $\text{ReLU}(x) = \max(0, x)$. Although a sigmoid activation is standard during training, for post-hoc interpretability we take the output-layer activation to be the threshold step function $\text{step}(\mathbf{z}) = 1$ if $\mathbf{z} \geq 0$ and $\text{step}(\mathbf{z}) = 0$ otherwise.

Tree Ensembles. Numerous ensemble methods exist; since our concern is post-hoc explanation, we specify the *inference* rule rather than the training procedure. Our analysis targets ensemble families whose inference uses either *majority voting* (e.g., random forests) or *weighted voting* (e.g., XGBoost). As with neural networks, however, our *tractability* results — the polynomial-time algorithms — extend to any ensemble whose inference is computable in polynomial time, covering a still broader range of ensemble configurations.

Majority Voting Inference. Under *majority voting*, the prediction $f(\mathbf{x})$ is the class $j \in [c]$ receiving the plurality of votes among the base predictions $f_i(\mathbf{x})$ over all $i \in [k]$ (i.e., over the trees in the ensemble):

$$f(\mathbf{x}) := \text{majority}(\{f_i(\mathbf{x}) \mid i \in [k]\}) \quad (10)$$

where $\text{majority}(S)$ denotes the most frequent label in the multiset S , and ties are broken by a fixed rule (e.g., lexicographic order or a predefined priority).

Weighted Voting Inference. Under *weighted voting*, each base model is assigned a weight $\phi_i \in \mathbb{Q}$ encoding its relative importance, and the prediction is the class of maximum total weight. Formally, for any $\mathbf{x} \in \mathbb{F}$:

$$f(\mathbf{x}) := \arg \max_{j \in [c]} \sum_{i=1}^k \phi_i \cdot \mathbb{I}[f_i(\mathbf{x}) = j] \quad (11)$$

where $\mathbb{I}$ denotes the identity function.

B.2 DISTRIBUTION FORMALIZATIONS

This subsection provides the formal definitions of the distribution classes introduced in the main text.

Empirical Distributions. An empirical distribution $\mathcal{D}$ over the input features is induced by a dataset $\mathbf{D}$ of inputs $\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}$, assigning to each $\mathbf{x} \in \mathbb{F}$ a probability equal to its empirical frequency in $\mathbf{D}$:

$$\mathbf{Pr}(\mathbf{x}) := \frac{1}{|\mathbf{D}|} \sum_{i=1}^{|\mathbf{D}|} \mathbb{I}(\mathbf{z}^i = \mathbf{x}) \quad (12)$$

Independent Distributions. Given a per-feature marginal probability $p(\mathbf{x}_i) \in [0, 1]$ for each coordinate, the distribution $\mathcal{D}$ is independent if its joint probability factorizes as $\mathbf{Pr}(\mathbf{x}) := \prod_{i \in [n]} p(\mathbf{x}_i)$.

When $p(\mathbf{x}_i) = p(\mathbf{x}_j)$ for all $i, j \in [n]$, this specializes to the *uniform distribution*, itself a special case of an independent distribution.

General Distributions. Many of our proofs — in particular those establishing intrinsic properties of the value functions — hold under general distributional assumptions. While empirical distributions (the training dataset as a proxy) are the most common choice in XAI, alternative distribution estimators include k-NN resampling from nearby points (Li et al., 2023; Almanjahie et al., 2018; Gweon et al., 2019), copula models of tabular dependencies (Größer & Okhrin, 2022), and expressive conditional generative models such as CTGAN (Xu et al., 2019) and conditional diffusion models (Huang et al., 2022). Such estimators are standard in the counterfactual-explanation and algorithmic-recourse literature (Karimi et al., 2021; Fokkema et al., 2024; 2023; Verma et al., 2024), which is closely related to contrastive explanations. Finally, when the value function is computed neither from a structural property of the model (e.g., leaf enumeration in decision trees) nor from an empirical distribution, it may be estimated via Monte Carlo sampling (Hastings, 1970), following approaches such as (Subercaseaux et al., 2025; Lopardo et al., 2023).

B.3 SUBSET VS. CARDINAL MINIMALITY

This subsection expands on the distinction between cardinal and subset minimality. Cardinal minimality is the strictly stronger guarantee: it constrains the *global* minimum explanation size, whereas subset minimality enforces only a *local* minimality condition.

For the forward implication, suppose $S \subseteq [n]$ is a cardinally minimal explanation. Then no set S' with $|S'| < |S|$ is an explanation; in particular no strict subset $S' \subsetneq S$ is an explanation, so S is subset-minimal.

The converse fails. Consider the function:

$$f := \mathbf{x}_1 \vee (\mathbf{x}_2 \wedge \mathbf{x}_3) \quad (13)$$

together with the assignment $\mathbf{x} := (1, 1, 1)$, for which $f(1, 1, 1) = 1$. Fixing the single feature $\mathbf{x}_1 = 1$ yields a cardinally minimal (hence subset-minimal) sufficient reason, since the output stays 1 for all values of $\mathbf{x}_2, \mathbf{x}_3$. Fixing instead both $\mathbf{x}_2 = 1$ and $\mathbf{x}_3 = 1$ yields a subset-minimal explanation that is *not* cardinally minimal. Thus every cardinally minimal explanation is subset-minimal, but not conversely.

C PROOF OF PROPOSITION 1

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Proof. It is known that certain Boolean functions admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons for some input $\mathbf{x}$ Bassan et al. (2024), a phenomenon exhibited by functions of the following form. We show that the same function admits an exponential number of *global probabilistic* sufficient reasons. This function is a special case of the broader class of *threshold Boolean functions* of Wegener (2005): for $n = 2k + 1$ with $k \in \mathbb{N}$, it is defined by

$$f(\mathbf{x}) := \begin{cases} 1 & \text{if } \sum_{i=1}^n \mathbf{x}_i \geq k + 1 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where $\mathbf{x}$ is drawn from a uniform distribution $\mathcal{D}$. The function f is *symmetric* in the sense of symmetric threshold Boolean functions Wegener (2005): its output depends only on the number of 1's (equivalently, 0's) in the input and is invariant under permutations of the input bits. Moreover, the two extreme inputs — the all-zeros vector $(0, \dots, 0)$ and the all-ones vector $(1, \dots, 1)$ — each admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons.

By symmetry of f , the equality $v_{\text{suff}}^g(S_1) = v_{\text{suff}}^g(S_2)$ holds for any two subsets $S_1, S_2 \subseteq [n]$. Moreover, for any subset $S \subseteq [n]$ of size $k + 1$ one verifies that:

$$(\forall i \in S). \quad v_{\text{suff}}^g(S) > v_{\text{suff}}^g(S \setminus \{i\}) \quad (15)$$

We further fix δ to satisfy:

$$v_{\text{suff}}^g(S \setminus \{i\}) < \delta \leq v_{\text{suff}}^g(S), \quad (16)$$

and set $k := \lfloor \frac{n}{2} \rfloor$. Each such subset is *subset-minimal*, since no subset of size at most $\lfloor \frac{n}{2} \rfloor - 1$ is a sufficient reason for $\langle f, \mathbf{x} \rangle$ under the δ threshold. The analysis of (Bassan et al., 2024) then applies directly, yielding exactly $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ subset-minimal local sufficient reasons for $\langle f, \mathbf{x} \rangle$. By Stirling's approximation:

$$\lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi}}{e^2} \cdot \frac{2^n}{\sqrt{n}} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor} \leq \frac{e}{\pi} \cdot \frac{2^n}{\sqrt{n}} \quad (17)$$

This gives the stated $\Theta(\frac{2^n}{\sqrt{n}})$ bound on the number of subset-minimal global δ -sufficient reasons.

□

D PROOF OF PROPOSITION 2

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Proof. We establish monotonicity for both families of global explanations, sufficient and contrastive, proceeding by successive generalization. We first prove the property for global sufficiency in the base case of Boolean functions, then lift it to discrete multi-valued input domains with multiple output classes, and subsequently to continuous input domains, thereby showing that monotonicity holds for every well-defined classification function. We then transfer the guarantee from global sufficient to global contrastive explanations. Crucially, no step of the argument invokes feature independence: monotonicity holds for *any* underlying distribution.

Finally, we exhibit an explicit counterexample witnessing that monotonicity fails for the local probabilistic sufficient and contrastive value functions.

Lemma 1. *The global sufficient value function v_{diff}^g is monotonic non-decreasing for Boolean functions, under any data distribution.*

Proof. Given an arbitrary set $S \in [n]$, and a feature $i \notin S$, we have that:

$$\Pr(\mathbf{x}) = \Pr(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) = \Pr(\mathbf{x}_{\bar{S}} | \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \quad (18)$$

Consequently, this implies that:

$$\begin{aligned} \Pr(\mathbf{x}) &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} | \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_{S \cup \{i\}}) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} | \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_S, x_i) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} | \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(x_i | \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \end{aligned} \quad (19)$$

Moreover, for any two points $\mathbf{x}, \mathbf{x}' \sim \mathcal{D}$ such that $\mathbf{x}_S = \mathbf{x}'_S$ and $f(\mathbf{x}) = 1 - f(\mathbf{x}')$, it holds that:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S) = 1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}') | \mathbf{z}_S = \mathbf{x}'_S) \quad (20)$$

For notational economy, we occasionally suppress the subscript $\sim \mathcal{D}$ in the local probability expression $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)$, and write f^+ and f^- for the events $f(\mathbf{z}) = 1$ and $f(\mathbf{z}) = 0$, respectively.

We first derive a technical simplification of the probability term, and then use it to establish monotonicity.

Simplifying the probability term. Let $\mathcal{D}_S$ denote the marginal of $\mathcal{D}$ restricted to the coordinates in S . Fix a point $\mathbf{x} \sim \mathcal{D}$ and a set S , and consider the set of all $\mathbf{z} \sim \mathcal{D}$ satisfying $\mathbf{z}_S = \mathbf{x}_S$ and $f(\mathbf{z}) = b$ for a fixed $b \in \{0, 1\}$; every such $\mathbf{z}$ carries the identical local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b | \mathbf{z}_S = \mathbf{x}_S)$. Marginalizing over $x_{\bar{S}}$ across all $\mathbf{x} \sim \mathcal{D}$ that agree on x_S and on the output b yields $\Pr(f(\mathbf{x}) = b, \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b | \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S)$, whence:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b | \mathbf{z}_S = \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b | \mathbf{x}_S). \quad (21)$$

Thus, for $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)]$, we get that:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^+} \Pr(f^+, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ | \mathbf{z}_S = \mathbf{x}_S) + \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^-} \Pr(f^-, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^- | \mathbf{z}_S = \mathbf{x}_S) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ | \mathbf{z}_S = \mathbf{x}_S))^2 + (\Pr_{\mathbf{z} \sim \mathcal{D}}(f^- | \mathbf{z}_S = \mathbf{x}_S))^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ | \mathbf{z}_S = \mathbf{x}_S))^2 + (1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ | \mathbf{z}_S = \mathbf{x}_S))^2], \end{aligned} \quad (22)$$

From which it follows that:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \sum_{\mathbf{x}_{S \cup \{t\}} \sim \mathcal{D}_{S \cup \{t\}}} \Pr(\mathbf{x}_{S \cup \{t\}}) \cdot [(\Pr_{\mathbf{z}}(f^+ | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 + (1 - \Pr_{\mathbf{z}}(f^+ | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2] \\ &= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(x_t = 1, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ | \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2 + (1 - \Pr_{\mathbf{z}}(f^+ | \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2] + \\ &\quad \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(x_t = 0, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ | \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2 + (1 - \Pr_{\mathbf{z}}(f^+ | \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2] \end{aligned} \quad (23)$$

Proving monotonicity. Building on the preceding simplification, we now prove the monotonicity claim. We first fix additional notation: let $g_1^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1)$ and $g_0^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0)$, and let $P_1 := \Pr(z_t = 1 \mid \mathbf{x}_S)$ and $P_0 := \Pr(z_t = 0 \mid \mathbf{x}_S)$. Then, the following expectations $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ and $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})]$ can be simplified accordingly¹:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 + (1 - (P_1 \cdot g_1^+ + P_0 \cdot g_0^+))^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1]. \end{aligned} \quad (24)$$

As a result, we obtain the following:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot ((g_1^+)^2 + (1 - g_1^+)^2) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot ((g_0^+)^2 + (1 - g_0^+)^2) \\ &= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2P_1 \cdot ((g_1^+)^2 - g_1^+) + 2P_0 \cdot ((g_0^+)^2 - g_0^+) + 1] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1]. \end{aligned} \quad (25)$$

Let $\Delta_t(S, f)$ denote $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$. Without loss of generality, we assume that both P_1 and P_0 are positive. Otherwise, it can be verified that $\Delta_t(S, f) = 0$. Then, we get:

$$\begin{aligned} \Delta_t(S, f) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot [(P_1 - P_1^2)(g_1^+)^2 + (P_0 - P_0^2)(g_0^+)^2 - 2 \cdot P_1 \cdot P_0 \cdot g_1^+ \cdot g_0^+] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 \cdot P_0 \cdot (g_1^+ - g_0^+)^2 \\ &\geq 0. \end{aligned} \quad (26)$$

Here we have used the identities $P_1 - P_1^2 = P_1(1 - P_1) = P_1 P_0$ and, symmetrically, $P_0 - P_0^2 = P_0(1 - P_0) = P_0 P_1$. Since $\Delta_t(S, f) \geq 0$, the value function $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone non-decreasing, which completes the proof. $\square$

Lemma 2. *The global sufficient value function v_{diff}^g is monotonic non-decreasing for functions with discrete multi-valued input domains and multiple output classes, under any data distribution.*

¹Note that the notations P_1 , P_0 , g_1^+ , and g_0^+ —as well as all others used throughout the paper—are not fixed constants, but rather depend on the specific input $\mathbf{x}_S$. Formally, they should be written as $P_1(\mathbf{x}_S)$, $P_0(\mathbf{x}_S)$, and so on. However, for readability, we omit the explicit dependence on $\mathbf{x}_S$.

Proof. We now lift the argument from the Boolean base case to functions whose input domains and output ranges are arbitrary finite sets. Let $\mathcal{K}$ denote the set of output classes.

We can generalize equation 22 as follows:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] \quad (27)$$

Given the condition $\mathbf{z}_S = \mathbf{x}_S$, suppose feature t can take r possible values $\{v_1, \dots, v_r\}$. Then, equation 23 can be generalized as follows:

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \sum_{k=1}^r \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = v_k}} \Pr(\mathbf{x}_S) \cdot \Pr(z_t = v_k \mid \mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k))^2 \right] \right]. \end{aligned} \quad (28)$$

Let P_k denote $\Pr(z_t = v_k \mid \mathbf{x}_S)$, Let g_k^j denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k)$, that is, z_t takes the value v_k and the output class is j . Then, we have

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \quad (29)$$

and

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) \right]$$

Combining these two implications, we can conclude that:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r (P_k \cdot g_k^j)^2 + \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right) \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \sum_{k \neq l} P_k \cdot P_l \cdot (g_k^j)^2 - \sum_{j \in \mathcal{K}} \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k \cdot P_l \cdot (g_k^j - g_l^j)^2 \right] \\ &\geq 0. \end{aligned} \quad (30)$$

Since $\Delta_t(S, f) \geq 0$, the value function $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone non-decreasing, which completes the proof in this setting. $\square$

Lemma 3. The global sufficient value function v_{suff}^g is monotonic non-decreasing for functions with continuous input domains and multiple output classes, under any data distribution.

Proof. We now lift the monotonicity result to features defined over continuous domains. One route is to partition each continuous feature domain into two disjoint subdomains and adapt the argument of lemma 2 with the corresponding change of notation; because both summation and integration are

linear, the reasoning carries over verbatim from the discrete to the continuous case, and only the notation changes.

For completeness, we instead give a direct proof. Assume all features range over continuous domains, and let $p(\mathbf{x}_S, \mathbf{x}_{\bar{S}})$ denote the joint probability density. Let $p(x_i)$ denote the marginal density of x_i , and $\mathcal{X}_i$ the domain of feature i .

We define marginal density as

$$p(\mathbf{x}_S) := \int_{\mathbf{x}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} p(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) d\mathbf{x}_{\bar{S}}, \quad (31)$$

where $\mathcal{X}_{\bar{S}}$ represents the domain of features in $\bar{S}$.

And the conditional density is defined as:

$$p(x_t | \mathbf{x}_S) := \frac{p(\mathbf{x}_S, x_t)}{p(\mathbf{x}_S)}. \quad (32)$$

The local probability of S given $\mathbf{x}$ is defined as

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \frac{\int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot p(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) d\mathbf{z}_{\bar{S}}}{p(\mathbf{x}_S)}, \quad (33)$$

This term can also be decomposed into:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \int_{z_t \in \mathcal{X}_t} p(z_t | \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \quad (34)$$

via an additional feature $t \notin S$.

We then modify equation 27 to

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_S = \mathbf{x}_S)] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S))^2 \right] d\mathbf{x}_S, \end{aligned} \quad (35)$$

while equation 28 is changed to

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p(z_t | \mathbf{x}_S) \cdot \left(\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S. \end{aligned} \quad (36)$$

For simplicity, let $Y_j(t)$ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_t)$, and $p_{t|S}$ denote $p(z_t \mid \mathbf{x}_S)$. Note that both $Y_j(t)$ and $p_{t|S}$ are associated with a fixed $\mathbf{x}_S$ ². We prove monotonicity as follows:

$$\begin{aligned}
\Delta_t(S, f) &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \left(\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S - \\
&\quad \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \right)^2 \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t)^2 dz_t - \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t) dz_t \right)^2 \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{2}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t)^2 - Y_j(t)Y_j(t')) dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&\geq 0.
\end{aligned} \tag{37}$$

□

Proposition 2. While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.

Proof. Combining lemmas 1 to 3, the global sufficient value function v_{suff}^g is monotone non-decreasing for every well-defined classification function — with Boolean, discrete, continuous, or mixed feature domains — under any data distribution.

$$v_{\text{con}}^g(S) = 1 - v_{\text{suff}}^g(\bar{S}) \tag{38}$$

Consequently v_{con}^g is monotone non-increasing in the features added to $\bar{S}$ or, equivalently, monotone non-decreasing as features are transferred from $\bar{S}$ to S , which completes the proof.

□

E PROOF OF PROPOSITION 3

Proposition 3. While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.

Proof. We establish the supermodularity property for both global sufficient and contrastive explanations. As in the monotonicity argument, we proceed by successive generalization — from Boolean functions to multi-valued discrete and then continuous input domains, and finally to arbitrary well-defined classification functions. In contrast to monotonicity, however, this argument requires the feature-independence assumption; we accordingly exhibit a counterexample showing that supermodularity can fail once independence is dropped.

Lemma 4. The global sufficient value function v_{suff}^g is supermodular for Boolean functions, under the assumption of feature independence.

²Changing $\mathbf{x}_S$ alters their values, we omit $\mathbf{x}_S$ in the notation for brevity.

Proof. Specifically, we prove that:

$$\Delta_t(S, f) \leq \Delta_t(S', f), \quad (39)$$

for all $S \subseteq S'$ with $t \notin S'$; that is, the marginal gain $\Delta_t(\cdot, f)$ is monotone in the conditioning set. The case $S' = S$ is immediate. It therefore suffices to treat $S' = S \cup \{i\}$ with $i \notin S$ and $i \neq t$. Under feature independence, the local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$ decomposes as:

$$\begin{aligned} & \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \\ &= \Pr(z_i = 1) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1) + \\ & \quad \Pr(z_i = 1) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0) + \\ & \quad \Pr(z_i = 0) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1) + \\ & \quad \Pr(z_i = 0) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0). \end{aligned} \quad (40)$$

Let P_1, P_0, Q_1 , and Q_0 denote $\Pr(z_t = 1), \Pr(z_t = 0), \Pr(z_i = 1)$, and $\Pr(z_i = 0)$, respectively. Moreover, let $g_{11}^+, g_{10}^+, g_{01}^+$, and g_{00}^+ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1)$, and $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0)$, respectively.

We further decompose equation 26 into:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [(Q_1 \cdot g_{11}^+ + Q_0 \cdot g_{01}^+) - (Q_1 \cdot g_{10}^+ + Q_0 \cdot g_{00}^+)]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+) + Q_0 \cdot (g_{01}^+ - g_{00}^+)]^2. \end{aligned} \quad (41)$$

Similarly, the following also holds:

$$\Delta_t(S \cup \{i\}, f) = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+)^2 + Q_0 \cdot (g_{01}^+ - g_{00}^+)^2]. \quad (42)$$

Thus, we get

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot Q_1 Q_0 \cdot (g_{11}^+ - g_{10}^+ - g_{01}^+ + g_{00}^+)^2 \\ &\geq 0. \end{aligned} \quad (43)$$

Hence $\Delta_t(\cdot, f)$ is monotone, i.e., $c(S \cup \{t\}, f) - c(S, f) \leq c(S' \cup \{t\}, f) - c(S', f)$, establishing supermodularity. $\square$

Lemma 5. *The global sufficient value function v_{suff}^g is supermodular for functions with discrete multi-valued input domains and multiple output classes, under the assumption of feature independence.*

Proof. We lift the previous result to multi-valued discrete functions, again under feature independence. Let P_k denote $\Pr(z_t = v_k)$ and $Q_{k'}$ denote $\Pr(z_i = v_{k'})$. Moreover, we use $g_{k'k}^j$ to denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = v_{k'}, z_t = v_k)$. We decompose equation 30 to

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} P_k P_l \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'k}^j - \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'l}^j \right]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} P_k P_l \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j) \right]^2. \end{aligned} \quad (44)$$

Likewise, we have

$$\begin{aligned}
\Delta_t(S \cup \{i\}, f) &= \sum_{k'=1}^{r'} \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_i = v'_k}} \mathbf{Pr}(\mathbf{x}_S) \cdot \mathbf{Pr}(z_i = v'_k) \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right].
\end{aligned} \tag{45}$$

Therefore, we get

$$\begin{aligned}
&\Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} \sum_{k' < l'} P_k P_l Q_{k'} Q_{l'} \cdot (g_{k'k}^j - g_{k'l}^j - g_{l'k}^j + g_{l'l}^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{46}$$

□

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{diff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{diff}^g is supermodular.*

Proof. We can generalize the supermodularity results established in Lemma 5 to the continuous case, assuming that $\mathcal{D}_p$ satisfies feature independence. One route is to partition each continuous feature domain into two disjoint subdomains and adapt the argument of Lemma 5 with the corresponding change of notation.

For completeness, we instead give a direct proof. Under feature independence, the marginal and conditional densities simplify to:

$$p(\mathbf{x}_S) = \prod_{i \in S} p(x_i), \text{ and } p(x_t | \mathbf{x}_S) = p(x_t). \tag{47}$$

Consequently, the local probability of S conditioned on $\mathbf{x}$ becomes:

$$\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot \prod_{i \in \bar{S}} p(z_i) d\mathbf{z}_{\bar{S}}. \tag{48}$$

We let $Y_j(i, t)$ denote $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_i = x_i, z_t = x_t)$, where $i \notin S, t \notin S$ and $i \neq t$. Evidently,

$$Y_j(i, t) = Y_j(t, i), \text{ and } Y_j(t) = \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i.$$

We have:

$$\begin{aligned}
&\Delta_t(S, f) \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i - \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t') dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S,
\end{aligned} \tag{49}$$

where

$$\begin{aligned} & \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \\ &= \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} \cdot (Y_j(i, t) - Y_j(i, t')) \cdot (Y_j(i', t) - Y_j(i', t')) dz_i dz_{i'} dz_t dz_{t'}. \end{aligned}$$

Let $S' = S \cup \{i\}$, then we have:

$$\begin{aligned} & \Delta_t(S', f) \\ &= \int_{\mathbf{x}_{S'} \in \mathcal{X}_{S'}} p(\mathbf{x}_{S'}) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_{S'} \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \int_{\mathcal{X}_i} p_i \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) dz_i \right] d\mathbf{x}_S \quad (50) \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \right) \right] d\mathbf{x}_S, \end{aligned}$$

where

$$\begin{aligned} & \int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \\ &= \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_{i'} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i dz_{i'}. \end{aligned}$$

Let $\alpha(i, t, t')$ denote $Y_j(i, t) - Y_j(i, t')$ and $\beta(i', t, t')$ denote $Y_j(i', t) - Y_j(i', t')$, we can infer that

$$\begin{aligned} & \Delta_t(S', f) - \Delta_t(S, f) \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \left[\sum_{j \in \mathcal{K}} \frac{1}{4} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} (\alpha(i, t, t') - \beta(i', t, t'))^2 dz_i dz_{i'} dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (51)$$

Since $\Delta_t(S', f) - \Delta_t(S, f) \geq 0$, the global sufficient value function v_{suff}^g is supermodular for every well-defined classification function, with Boolean, discrete, continuous, or mixed feature domains.

We now show that the local probabilistic sufficient value function is neither submodular nor supermodular. It suffices to exhibit a counterexample: fix the uniform distribution over the three variables $\{x_1, x_2, x_3\}$ and define the Boolean function

$$f(x_1, x_2) = (x_1 \vee x_2) \wedge x_3. \quad (52)$$

Take the input $\mathbf{x} = (1, 1, 1)$, which is classified as 1. The local probabilistic sufficient value function v_{suff}^l then evaluates to:

$$\begin{aligned} v_{\text{suff}}^l(\emptyset) &= \frac{3}{8}, \quad v_{\text{suff}}^l(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{3\}) = \frac{3}{4}, \\ v_{\text{suff}}^l(\{1, 2\}) &= \frac{1}{2}, \quad v_{\text{suff}}^l(\{1, 3\}) = 1, \quad v_{\text{suff}}^l(\{2, 3\}) = 1, \quad v_{\text{suff}}^l(\{1, 2, 3\}) = 1. \end{aligned}$$

Since $(v_{\text{suff}}^l(\{1, 2\}) - v_{\text{suff}}^l(\{2\})) < (v_{\text{suff}}^l(\{1\}) - v_{\text{suff}}^l(\emptyset)) < (v_{\text{suff}}^l(\{1, 3\}) - v_{\text{suff}}^l(\{3\}))$, the function v_{suff}^l is neither supermodular nor submodular. □

F PROOF OF PROPOSITION 4

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^l is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

Proof. Given the supermodularity of the global sufficient setting, submodularity of the global contrastive setting follows by duality. Let $S \subseteq [n]$ be the set of features permitted to change, and suppose $S \subseteq S'$ with $i \notin S'$. Comparing the marginal contributions $v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)$ and $v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')$ is equivalent to comparing $(1 - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}))$ and $(1 - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}'))$. Since $S \subseteq S'$ implies $\bar{S}' \subseteq \bar{S}$, supermodularity of v_{suff}^g yields:

$$(v_{\text{suff}}^g(\bar{S}) - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) \geq (v_{\text{suff}}^g(\bar{S}') - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) \quad (53)$$

Which is equivalent to:

$$(v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)) \geq (v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')). \quad (54)$$

Thus the global contrastive value function v_{con}^g is submodular, which completes the proof.

We now show that the local probabilistic contrastive value function is neither submodular nor supermodular. It suffices to exhibit a counterexample: fix the uniform distribution over the three variables $\{x_1, x_2, x_3\}$ and define the Boolean function

$$f(x_1, x_2) := (x_1 \vee x_2) \wedge x_3. \quad (55)$$

Take the input $\mathbf{x} = (1, 1, 1)$, which is classified as 1. The local probabilistic contrastive value function v_{con}^l then evaluates to:

$$\begin{aligned} v_{\text{con}}^l(\emptyset) &= 1, \quad v_{\text{con}}^l(\{1\}) = 1, \quad v_{\text{con}}^l(\{2\}) = 1, \quad v_{\text{con}}^l(\{3\}) = \frac{1}{2}, \\ v_{\text{con}}^l(\{1, 2\}) &= \frac{3}{4}, \quad v_{\text{con}}^l(\{1, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{2, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{1, 2, 3\}) = \frac{3}{8}. \end{aligned}$$

Since $(v_{\text{con}}^l(\{1, 2\}) - v_{\text{con}}^l(\{2\})) < (v_{\text{con}}^l(\{1, 2, 3\}) - v_{\text{con}}^l(\{2, 3\})) < (v_{\text{con}}^l(\{1\}) - v_{\text{con}}^l(\emptyset))$, the function v_{con}^l is neither submodular nor supermodular. □

In the remainder of this section, we show that feature independence is not merely sufficient but also necessary for supermodularity, via the following counterexample.

Define the Boolean function:

$$f(x_1, x_2) := x_1 \quad (56)$$

so that $f(00) = 0$, $f(01) = 0$, $f(10) = 1$, and $f(11) = 1$, and let $\mathcal{D}$ be an arbitrary distribution over (x_1, x_2) . Recall that the global sufficient value function is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x} \sim \mathcal{D}} \mathbf{Pr}(\mathbf{x}) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$$

and that supermodularity is equivalent to the inequality

$$\Delta_t(S, f) \leq \Delta_t(S', f),$$

holding for all $S \subseteq S'$ with $t \notin S'$. For $S = \emptyset$:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \\ &\quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) \end{aligned}$$

Which is also equivalent to:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)) \\
& = (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1))^2 + (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1))^2
\end{aligned}$$

For $S = \{2\}$:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 1) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 0) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 1)
\end{aligned}$$

This can also be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \\
& = \frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)}
\end{aligned}$$

In contrast, for $S = \{1\}$:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1)
\end{aligned}$$

This can similarly be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} = 1
\end{aligned}$$

Finally, for $S = \{1, 2\}$:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 0) + \\
&\quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 1) + \\
&\quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 0) + \\
&\quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 1)
\end{aligned}$$

This can likewise be reordered as follows:

$$\begin{aligned}
& \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0)} + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1)} + \\
& \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 0)}{\mathbf{Pr}(x_1 = 1, x_2 = 0)} + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 1)}{\mathbf{Pr}(x_1 = 1, x_2 = 1)} \\
&= 1
\end{aligned}$$

Set $t = 2$. Then $\Delta_t(\{1\}, f) = 0$. We next analyze $\Delta_t(\emptyset, f)$. Under feature independence:

$$\begin{aligned}
& (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1))^2 + (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1))^2 \\
&= \mathbf{Pr}(x_1 = 0)^2 + \mathbf{Pr}(x_1 = 1)^2,
\end{aligned}$$

and, similarly:

$$\begin{aligned}
& \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 0)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 1)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \\
&= \mathbf{Pr}(x_1 = 0)^2 + \mathbf{Pr}(x_1 = 1)^2.
\end{aligned}$$

Hence $\Delta_t(\emptyset, f) = 0$, so that $\Delta_t(\emptyset, f) \leq \Delta_t(\{1\}, f)$, as required for supermodularity. Absent feature independence, however, $\Delta_t(\emptyset, f) = 0$ need not hold. To see this, set:

$$\mathbf{Pr}(x_1 = 0, x_2 = 0) = P_{00}, \mathbf{Pr}(x_1 = 0, x_2 = 1) = P_{01}$$

and

$$\mathbf{Pr}(x_1 = 1, x_2 = 0) = P_{10}, \mathbf{Pr}(x_1 = 1, x_2 = 1) = P_{11}$$

Simplifying $\Delta_t(\emptyset, f)$ then gives:

$$\Delta_t(\emptyset, f) = \left[\frac{P_{00}^2 + P_{10}^2}{P_{00} + P_{10}} + \frac{P_{01}^2 + P_{11}^2}{P_{01} + P_{11}} \right] - \left[(P_{00} + P_{01})^2 + (P_{10} + P_{11})^2 \right] \quad (57)$$

Instantiating $P_{00} = 0.4$, $P_{01} = 0.2$, $P_{10} = 0.1$, and $P_{11} = 0.3$ yields:

$$\Delta_t(\emptyset, f) = \left[\frac{(0.4)^2 + (0.1)^2}{0.5} + \frac{(0.2)^2 + (0.3)^2}{0.5} \right] - \left[(0.4 + 0.2)^2 + (0.1 + 0.3)^2 \right] = 0.08. \quad (58)$$

Thus $\Delta_t(\emptyset, f) > \Delta_t(1, f)$, violating supermodularity. This establishes that — unlike monotonicity, which requires no such assumption — feature independence is a necessary condition for supermodularity.

G PROOF OF PROPOSITION 5

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Proof. The proposition follows directly from the monotonicity of both v_{suff}^g and v_{con}^g (Proposition 2), together with established results from prior work. We now make the reduction explicit.

Convergence of Algorithm 1 to a subset-minimal global δ -sufficient or δ -contrastive reason with respect to v_{con}^g and v_{suff}^g is an immediate consequence of monotonicity: prior work (Ignatiev et al., 2019; Wu et al., 2024b; Arenas et al., 2022) shows that this class of greedy elimination procedures returns a subset-minimal explanation whenever the value function is monotone non-decreasing. At termination, $v(S \setminus \{i\}) < \delta$ for every $i \in S$, while the invariant $v(S) \geq \delta$ is maintained; that is, no single deletion preserves feasibility. Monotonicity extends this from single deletions to all proper subsets, so that no proper subset of S is feasible and S is therefore subset-minimal.

We next show that Algorithm 1 fails to converge to a subset-minimal solution in the local setting. It is well established that the local probabilistic sufficient value function (and, symmetrically, the local probabilistic contrastive value function) is non-monotone Arenas et al. (2022); Izza et al. (2023); Subercaseaux et al. (2025); Izza et al. (2024b). It suffices to exhibit a counterexample: fix the uniform distribution over the two binary variables $\{x_1, x_2\}$ and define the Boolean function

$$f(x_1, x_2) := x_1 \vee x_2. \quad (59)$$

Take the input $\mathbf{x} = (0, 1)$, which is classified as 1; then:

$$v_{\text{suff}}^{\ell}(\emptyset) = \frac{3}{4}, \quad v_{\text{suff}}^{\ell}(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^{\ell}(\{2\}) = 1, \quad v_{\text{suff}}^{\ell}(\{1, 2\}) = 1. \quad (60)$$

Since $v_{\text{suff}}^{\ell}(\emptyset) > v_{\text{suff}}^{\ell}(\{1\}) < v_{\text{suff}}^{\ell}(\{1, 2\})$, the function v_{suff}^{ℓ} is non-monotone. Symmetrically, one computes:

$$v_{\text{con}}^{\ell}(\emptyset) = 1, \quad v_{\text{con}}^{\ell}(\{1\}) = 1, \quad v_{\text{con}}^{\ell}(\{2\}) = \frac{1}{2}, \quad v_{\text{con}}^{\ell}(\{1, 2\}) = \frac{3}{4}. \quad (61)$$

(For v_{con}^{ℓ} , S denotes the set of features permitted to vary.) Since $v_{\text{con}}^{\ell}(\emptyset) > v_{\text{con}}^{\ell}(\{2\}) < v_{\text{con}}^{\ell}(\{1, 2\})$, the function v_{con}^{ℓ} is likewise non-monotone.

□

H PROOF OF THEOREM 1

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Proof. We first show that, for decision trees under an independent distribution $\mathcal{D}$, the value function

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (62)$$

is computable in polynomial time, by the following procedure. Exploiting the tractable structure of decision trees, we enumerate all pairs of leaves, corresponding to partial assignments $\mathbf{x}_S$ and $\mathbf{z}_{S'}$. Whenever the two leaves induce the same prediction under f , we evaluate the associated term of $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ as the product of the feature-wise probabilities over the coordinates shared by the two vectors. Feature independence makes these probabilities factorize, so the full expectation is the sum of the contributions over all matching leaf pairs.

Because every probability is supplied in the input and each operation is polynomial-time, the entire procedure runs in polynomial time. Hence every step of Algorithm 1 is polynomial-time, and so is the algorithm as a whole. Combining this with Lemma 2, which establishes that v_{suff}^g is monotone non-decreasing, Algorithm 1 converges to a subset-minimal explanation: at termination $v(S \setminus \{i\}) < \delta$ for all $i \in S$ while the invariant $v(S) \geq \delta$ is preserved, and monotonicity extends this to all proper subsets, so S is subset-minimal.

The remaining part of the claim follows from (Arenas et al., 2022), which proved that, unless $P=NP$, no polynomial-time algorithm computes a subset-minimal local δ -sufficient reason for decision trees. Although established under the uniform distribution, this lower bound extends to independent distributions, which subsume the uniform case. Together, the two parts yield the local/global complexity separation, completing the proof.

□

I PROOF OF PROPOSITION 6

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

Proof. The computation of the global sufficient probability function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (63)$$

or the computation of the global contrastive probability function

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (64)$$

is computable in polynomial time whenever $\mathcal{D}$ is an empirical distribution: it suffices to enumerate the pairs of instances $\mathbf{x}, \mathbf{z}$ in the dataset and evaluate $f(\mathbf{x})$ and $f(\mathbf{z})$ by inference, accumulating the expectation term-by-term. Consequently, testing whether the value function meets the threshold δ is also polynomial-time. Invoking the non-decreasing monotonicity of the global value functions (Proposition D), Algorithm 1 applies to both the contrastive and the sufficient search. Since it issues only a linear number of polynomial-time value queries, a subset-minimal explanation — contrastive or sufficient — is obtained in polynomial time.

□

J PROOF OF THEOREM 2

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $PTIME=NP$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

Proof. The first part follows directly from Proposition 6, which holds for any model, since both $v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ and $v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ are computable in polynomial time under an empirical distribution. For the second part, we establish two lemmas: one for the intractability of the sufficient case, the other for the contrastive case.

Lemma 6. *Unless $PTIME=NP$, then there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We prove the claim first for neural networks and then extend it to tree ensembles. The argument is a polynomial-time reduction: we show that a polynomial-time algorithm for computing a subset-minimal local δ -sufficient reason for neural networks would decide the NP-complete *CNF-SAT* problem in polynomial time. *CNF-SAT* is defined as follows:

CNF-SAT:

Input: A formula in conjunctive normal form (CNF): ϕ .

Output: Yes, if there exists an assignment to the n literals of ϕ such that ϕ is evaluated to True, and No otherwise

The reduction additionally relies on the following lemma (proved in Barceló et al. (2020)):

Lemma 7. Any boolean circuit ϕ can be encoded into an equivalent MLP over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.

Proof. We establish hardness already for a restricted special case, which a fortiori implies hardness in the general setting. Specifically, we take the empirical distribution $\mathcal{D}$ to consist of a single point, which for concreteness we fix to be the zero vector $\mathbf{0}_n$. Since the sufficiency of S then depends only on the local instance $\mathbf{x}$ and the single support point $\mathbf{0}_n$, the input format (discrete, continuous, etc.) is immaterial, and the hardness result holds uniformly across all such formats.

Let $\mathbf{1}_n$ denote the n -dimensional all-ones vector. Given an input ϕ , we first evaluate $\phi(\mathbf{1}_n)$, i.e., assign all variables to True; if $\phi(\mathbf{1}_n) = 1$ then ϕ is satisfiable and we are done, so we may assume $\phi(\mathbf{1}_n) = 0$. Define:

$$\begin{aligned}\phi_2 &:= (x_1 \wedge x_2 \wedge \dots \wedge x_n), \\ \phi' &:= \phi \vee \phi_2\end{aligned}\tag{65}$$

Although ϕ' is no longer a CNF, Lemma 8 still encodes it into an MLP f that agrees with ϕ' over $\{0, 1\}^n$. Under the assumption that a subset-minimal local δ -sufficient reason is computable in polynomial time, we compute one for the instance $\langle f, \mathbf{x} := \mathbf{1}_n \rangle$ with $\delta := 1$, where $\mathcal{D}$ is the empirical distribution supported on the single point $\mathbf{0}_n$.

We claim that the subset-minimal δ -sufficient reason for $\langle f, \mathbf{x} \rangle$ equals the full feature set, i.e., $S = \{1, \dots, n\}$, if and only if $\langle \phi \rangle \notin \text{CNF-SAT}$.

Suppose first that $S = \{1, \dots, n\}$. Because S is δ -sufficient for $\langle f, \mathbf{x} \rangle$, fixing the complementary set to any value preserves the prediction; as the complementary set is $\emptyset$, this holds trivially. Subset minimality of S means every $S' \subsetneq S$ satisfies $v(S') < \delta = 1$. Since $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$ is determined by the single support point $\mathbf{0}_n$, the value function is $\{0, 1\}$ -valued, whence $v(S') = 0$. Thus, for every $S' \subsetneq S$, fixing the features in S' to 1 and the rest to 0 does not yield the prediction 1. As the ϕ_2 conjunct of ϕ' is True only under the all-ones assignment, ϕ evaluates to False on all these inputs; combined with the earlier verification that $\phi(\mathbf{1}_n) = 0$, no assignment satisfies ϕ , so $\phi \notin \text{CNF-SAT}$.

Conversely, suppose $S \subsetneq \{1, \dots, n\}$. Since S is δ -sufficient under $\mathcal{D}$ with $\delta = 1$, the same reasoning gives $v(S) = 1$: fixing the features in S to 1 and the rest to 0 forces $f = 1$. As this assignment is not the all-ones vector, the ϕ_2 conjunct of ϕ' is False, so the ϕ conjunct must be True, whence $\langle \phi \rangle \in \text{CNF-SAT}$. This completes the reduction.

The claim is thus established for neural networks. Extending it to tree ensembles requires only a minor adaptation, via the following lemma noted in prior work (Ordyniak et al., 2024; Audemard et al., 2022b):

Lemma 8. Any CNF or DNF ϕ can be encoded into an equivalent random forest classifier over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.

Proof. We reuse the construction from the neural network proof, which encoded ϕ' into an equivalent network. The construction rendered ϕ' non-CNF (which posed no obstacle for the MLP encoding, as any Boolean circuit admits one). However, since ϕ' is a conjunction of only two terms, it is readily rewritten as an equivalent CNF:

$$\phi' := (c_1 \vee \phi_2) \wedge (c_2 \vee \phi_2) \wedge \dots \wedge (c_m \vee \phi_2)\tag{66}$$

where each c_i is a disjunction of a few terms. Thus ϕ' is a valid CNF and, by the lemma, encodes into an equivalent random forest classifier; the reduction given for MLPs then transfers verbatim, completing the proof for both model families.

□

Lemma 9. Unless $\text{PTIME} = \text{NP}$, then there is no polynomial time algorithm for computing a subset minimal local δ -contrastive reason for either a tree ensemble or a neural network, under empirical distributions.

Proof. The argument is analogous to that of Lemma 6, again reducing from the NP-hard CNF-SAT problem defined in Lemma 6. Given a Boolean formula ϕ , we show that computing a subset-minimal contrastive reason — for a neural network or a tree ensemble — decides the satisfiability of ϕ .

We first test whether the all-ones assignment satisfies ϕ ; if so, ϕ is satisfiable and we are done. Otherwise, Lemma 8 encodes the CNF formula as a neural network f . As in Lemma 6, we set $\mathcal{D}$ to be the empirical distribution supported on the single point $\mathbf{0}_n$, fix $\delta = 1$, and compute a subset-minimal δ -contrastive reason for $\langle f, \mathbf{x} \rangle$.

We now show that a subset-minimal δ -contrastive reason exists if and only if ϕ is satisfiable, by reasoning parallel to Lemma 6. Since $\mathcal{D}$ is supported on a single point, the term $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ is $\{0, 1\}$ -valued. With $\delta = 1$, existence of a contrastive reason is equivalent to existence of an assignment that flips the classification of f — equivalently, that drives ϕ from False to True; such an assignment exists exactly when ϕ is satisfiable. Conversely, absence of any subset-minimal contrastive reason means no subset of features fixed to zero (with the complement set to ones) yields the prediction True, i.e., no assignment satisfies f (hence ϕ), so ϕ is unsatisfiable.

The extension from neural networks to tree ensembles follows exactly as in Lemma 6, encoding the CNF formula into an equivalent random forest classifier; the reduction remains valid, which completes the proof.

□

K PROOF OF THEOREM 3

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We decompose the proof into two lemmas: one establishing the approximation guarantee for the global setting, and one establishing the absence of any bounded approximation for the local setting.

Lemma 10. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence.*

Proof. We prove a stronger statement: the guarantee holds for *any* model satisfying the (mild) condition of polynomial-time inference, together with one further mild condition specified below — both of which are met by our neural network and tree ensemble formalizations.

Since we work with empirical distributions, the global contrastive value function

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (67)$$

is computable in polynomial time by enumerating all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as established in Proposition 2. Since Algorithm 2 issues only a linear number of these polynomial-time queries, its total runtime is polynomial.

For the approximation factor, the classical result of Wolsey et al. (Wolsey, 1982) gives a harmonic-series approximation guarantee for integer-valued monotone non-decreasing submodular functions v , and more generally an approximation factor of $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(\{i\})}\right)\right)$. Proposition 2 establishes that v_{con}^g is monotone non-decreasing, and Proposition 4 establishes its submodularity under feature independence. Together with the polynomial runtime of Algorithm 2, this yields the approximation factor $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$.

It remains to show that $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ is finite and bounded by $\mathcal{O}(\ln(|D|))$, which is constant for a fixed dataset $|D|$. We first ensure the expression is well-defined and finite, then derive the stated bound. The key device is a preprocessing step that removes the “redundant” features which could otherwise drive the denominator $\min_{i \in [n]} v_{\text{con}}^g(\{i\})$ to zero. We first define redundancy:

Definition 1. Let $\mathcal{D}$ denote some empirical distribution over a dataset D . Then we say that some feature $i \in [n]$ is redundant with respect to $\mathcal{D}$ if for any pair $\mathbf{x}, \mathbf{z} \in D$ it holds that $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) = f(\mathbf{x})$.

Here $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}})$ denotes the vector with all coordinates in $[n] \setminus \{i\}$ fixed to their values in $\mathbf{x}$ and coordinate i set to its value in $\mathbf{z}$. This condition is equivalent to:

$$\forall S \subseteq [n], \mathbf{z} \in D \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}_{S \setminus \{i\}}; \mathbf{z}_{S \cup \{i\}}) \quad (68)$$

As before, $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ denotes the vector with coordinates in S fixed to their values in $\mathbf{x}$ and coordinates in $\bar{S}$ fixed to their values in $\mathbf{z}$. Thus, deleting feature i from the input space and defining a reduced function f' over $[n'] := [n] \setminus \{i\}$ yields, for every input of dimension n' , an output of f' that coincides with that of f restricted to the coordinates $[n] \setminus \{i\}$. This deletion is polynomial-time for both model families: for neural networks, one detaches the corresponding input neuron; for tree ensembles, one removes every split on that feature across the constituent trees. Given the empirical structure of $v_{\text{con}}^g(S)$ — as noted earlier in this proof and in Proposition 2 — each $v_{\text{con}}^g(\{i\})$, for all $i \in [n]$, is computed by enumerating the pairs $\mathbf{x}, \mathbf{z} \in \mathcal{D}$ and testing whether:

$$f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) \neq f(\mathbf{x}) \quad (69)$$

If this holds for some pair, feature i is not redundant; if it fails for all pairs, i is redundant. After identifying all redundant features with respect to $\mathcal{D}$, we remove them and construct the equivalent model f' , which, as noted, is a polynomial-time transformation for both neural networks and decision trees.

To conclude, observe that $v_{\text{con}}^g(\{i\}) = 0$ under an empirical distribution $\mathcal{D}$ if and only if feature i is redundant with respect to $\mathcal{D}$. Hence, once the preprocessing step removes all redundant features and constructs f' , we have $\min_{i \in [n]} v_{\text{con}}^g(\{i\}) > 0$.

We now derive the more precise bound referenced above, namely that the approximation factor is $\mathcal{O}(\ln(|D|))$. Since the empirical distribution $\mathcal{D}$ is fixed, this is a constant. Tighter bounds are possible; our sole aim is to certify constancy, which will contrast sharply with the local setting, where no bounded-factor approximation exists.

Concretely, by the definition of empirical distributions the probability term $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ is at least $\frac{1}{|D|}$, so its expectation $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ (i.e., v_{con}^g) is at least $\frac{1}{|D|^2}$. In particular:

$$\min_{i \in [n]} v_{\text{con}}^g(\{i\}) \geq \frac{1}{|D|^2} \quad (70)$$

Consequently, we obtain that:

$$\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right) \leq \ln(|D|^2) = \mathcal{O}(\ln(|D|)) \quad (71)$$

which completes the proof.

□

Lemma 11. Unless $P\text{TIME} = NP$, there is no bounded approximation for computing a cardinally minimal local δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$ where f is either a neural network or a tree ensemble, even when $|D| = 1$.

Proof. Suppose, for contradiction, that a polynomial-time bounded-factor approximation algorithm exists for the cardinally minimal local δ -contrastive reason of some $\langle f, \mathbf{x} \rangle$, with f a neural network or tree ensemble, even when $|D| = 1$. Lemma 9 shows that even for a single baseline $\mathbf{z} = 0_n$ (a one-instance dataset), deciding the existence of a contrastive reason is NP-hard unless PTIME = NP. A bounded-factor approximation would, however, decide this existence question, since a finite approximation ratio can only be returned when a contrastive reason exists; this contradicts the lower bound.

□

L PROOF OF THEOREM 4

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{diff}}^g([n])}{\min_{i \in [n]} v_{\text{diff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless PTIME=NP, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. We decompose the proof into two lemmas: one establishing the approximation guarantee for the global setting, and one establishing the intractability of the local setting.

Lemma 12. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{diff}}^g([n])}{\min_{i \in [n]} v_{\text{diff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence.*

Proof. The proof will follow a similar approach to that of Lemma 10, where we showed that for both neural networks and tree ensembles, Algorithm 2 achieves an approximation factor of $\ln\left(\frac{v_{\text{diff}}^g([n])}{\min_{i \in [n]} v_{\text{diff}}^g(\{i\})}\right)$. After applying the preprocessing step, this ratio is guaranteed to be finite and bounded by $\mathcal{O}(\ln(|D|))$.

As in Lemma 10, we prove the stronger claim that the result holds for *any* model satisfying the (mild) polynomial-time inference condition together with the redundant-feature-removal condition described in Lemma 10; both are met by our neural network and tree ensemble formalizations.

Here too, since we work with empirical distributions, the global sufficient value function

$$v_{\text{diff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (72)$$

is computable in polynomial time by enumerating all pairs $\mathbf{x}, \mathbf{z}$ in the dataset D , as shown in Lemma 10 and Proposition 2. Since Algorithm 2 issues only a linear number of these polynomial-time queries, its overall runtime is polynomial.

Whereas Lemma 10 invoked the guarantee of Wolsey et al. (Wolsey, 1982) for monotone non-decreasing submodular functions, the supermodular structure here calls for a different guarantee. Specifically, Shi et al. (Shi et al., 2021) give an approximation factor of $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{diff}}^g([n])}{\min_{i \in [n]} v_{\text{diff}}^g(\{i\})}\right)\right)$, with total curvature $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$. Proposition 2 establishes that v_{diff}^g is monotone non-decreasing, and Proposition 3 establishes its supermodularity under feature independence. Combined with the polynomial runtime of Algorithm 2, this yields the approximation factor $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{diff}}^g([n])}{\min_{i \in [n]} v_{\text{diff}}^g(\{i\})}\right)\right)$, with $k^f := 1 - \min_{i \in [n]} \frac{v_{\text{diff}}^g([n]) - v_{\text{diff}}^g([n] \setminus \{i\})}{v_{\text{diff}}^g(\{i\}) - v_{\text{diff}}^g(\emptyset)}$.

To show this expression is bounded and constant, we apply the same preprocessing step as in Lemma 10, removing all redundant features from f — again in polynomial time for both neural networks and tree ensembles. As before, a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. The preprocessing yields a function f' that agrees with f over the remaining features and guarantees, for every i , both $v_{\text{diff}}^g(\{i\}) > 0$ and $v_{\text{diff}}^g(\{i\}) - v_{\text{diff}}^g(\emptyset) > 0$.

Equivalently, this preprocessing phase mirrors that of Lemma 10: it runs in polynomial time for both neural networks and tree ensembles, a feature i is strictly redundant exactly when $v(\{i\}) - v(\emptyset) = 0$,

and the resulting function f' both agrees with f over all remaining features and satisfies $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ for every i .

Following the same reasoning as in Lemma 10, where the probability term is lower-bounded by $\frac{1}{|D|}$ and its expectation by $\frac{1}{|D|^2}$, we obtain $\min_{i \in [n]} v_{\text{suff}}^g(i) \geq \frac{1}{|D|^2}$ and $v_{\text{suff}}^g(i) - v_{\text{suff}}^g(\emptyset) \geq \frac{1}{|D|^2}$. As in Lemma 10, the first term is then bounded by $\mathcal{O}(\ln(|D|))$; for the second term:

Since v_{suff}^g is supermodular (Proposition 3), we have $v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})$, whence $1 \leq v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\}) \leq \frac{1}{|D|^2}$. Therefore $1 \leq \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)} \leq |D|^2$, so $1 - |D|^2 \leq k^f \leq 0$, and consequently $\frac{1}{1-k^f} \leq |D|^2$. Hence the overall approximation factor of Shi et al. is constant. This factor may be substantially smaller in practice; our objective here is only to certify that it is constant — in sharp contrast to the local setting, where no bounded-factor approximation exists.

□

Lemma 13. *Unless $\text{PTIME} = \text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. The argument parallels Lemma 11. Suppose, for contradiction, that a polynomial-time bounded-factor approximation algorithm exists for the cardinally minimal local δ -sufficient reason of some $\langle f, \mathbf{x} \rangle$, with f a neural network or tree ensemble, even when $|D| = 1$. Lemma 6 shows that even for a single baseline $\mathbf{z} = 0_n$, deciding the existence of a sufficient reason is NP-hard unless $\text{PTIME} = \text{NP}$. Such an approximation algorithm would decide this existence question, contradicting the lower bound.

□

M HARDNESS OF APPROXIMATING GLOBAL CONTRASTIVE REASONS FOR NEURAL NETWORKS USING EMPIRICAL DISTRIBUTIONS

Proposition 1. *Assume a neural network $f : \mathbb{F} \rightarrow [c]$ with ReLU activations, some dataset $\mathbf{D}$, such that $\mathcal{D}$ is the empirical distribution defined over $\mathbf{D}$, and some threshold $\delta \in \mathbb{Q}$. Then unless $\text{NP} \subseteq \text{DTIME}(n^{\mathcal{O}(\log(\log(n)))})$, there does not exist a polynomial-time $\mathcal{O}(\ln(|D|) - \epsilon)$ approximation algorithm for computing a cardinally minimal global contrastive reason for f concerning $\mathcal{D}$, for any fixed $\epsilon > 0$.*

Proof. The proof is a polynomial-time approximation-preserving reduction from a variant of Set Cover that, like standard Set Cover, is provably hard to approximate within the factor stated in the proposition (Vazirani, 2001).

Partial Set Cover:

Input: A universe $U := \{u_1, \dots, u_n\}$, a collection of sets $S := \{S_1, \dots, S_m\}$ such that for all j $S_j \subseteq U$, and a coverage threshold $\delta \in \{0, 1, \dots, n\}$.

Output: A minimum-size subset $C \subseteq S$ such that $|\cup_{S_j \in C} S_j| \geq \delta$.

We reduce this problem to computing a *global*, cardinally minimal contrastive reason S for a neural network f with respect to a distribution $\mathcal{D}$ over a dataset $\mathbf{D}$ and a threshold δ' . We first recall the definition of a global contrastive reason and instantiate it for an empirical dataset $\mathbf{D}$:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (73)$$

Under the empirical distribution over $\mathbf{D}$, this expands to:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|D|} \sum_{\mathbf{x} \in \mathbf{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|D|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \right) & \end{aligned} \quad (74)$$

where $\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x}) := \{\mathbf{z} \in \mathbf{D} : \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}\}$. We construct a ReLU neural network f , first specifying the target functional form it is to implement. For each set S_j , introduce two vectors $\mathbf{x}^j, \mathbf{y}^j \in \mathbb{Z}^n$ from the dataset $\mathbf{D}$ that agree on all coordinates outside S_j , i.e., coincide on the complement $\bar{S}_j$:

$$\mathbf{x}_{\bar{S}_j}^j = \mathbf{y}_{\bar{S}_j}^j \quad (75)$$

and differ on S_j , with those coordinates drawn from two distinct vectors $\mathbf{z}$ and $\mathbf{z}'$:

$$\mathbf{x}^j = (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = (\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = \mathbf{y}^j \quad (76)$$

We require f to satisfy:

$$f(\mathbf{x}^j) = f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = f(\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}'_{S_j}) = f(\mathbf{y}^j) \quad (77)$$

Assume further that $\mathbf{x}^j$ and $\mathbf{y}^j$ each occur *uniquely* in $\mathbf{D}$ over all $[n]$ coordinates — no other vector of $\mathbf{D}$ matches either on any coordinate. This assumption yields:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}_j} = \mathbf{x}_{\bar{S}_j})] = \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{|\mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})|} \right) = \frac{2}{2|\mathbf{D}|} = \frac{1}{|\mathbf{D}|} \quad (78)$$

The factor 2 in both numerator and denominator arises because exactly two ordered vector pairs — $(\mathbf{x}^j, \mathbf{z}^j)$ and $(\mathbf{z}^j, \mathbf{x}^j)$ — contribute to the respective sums.

We repeat this construction for each S_i in the set-cover instance, each time selecting two *distinct* vectors $\mathbf{x}^i, \mathbf{y}^i$ that differ on *every* coordinate in $[n]$. This produces $2m$ vectors constituting the dataset $\mathbf{D}$ from which f is built.

We realize this target function by a neural network f via a known *neural network memorization* result (Vardi et al., 2022), which guarantees that a ReLU network representing any finite set of input–output pairs is constructible in polynomial time. We construct f accordingly and set $\delta' := \frac{\delta}{|\mathbf{D}|}$.

It remains to show that a subset S_j is a global contrastive reason for f under tolerance δ' if and only if C is a partial set cover for U under tolerance δ . By Equation 78, each constructed pair $(\mathbf{x}^j, \mathbf{y}^j)$ contributes exactly $\frac{1}{|\mathbf{D}|}$ to the value term. Since each S_j uses its own distinct pair of vectors in $\mathbb{Z}^n$, these contributions do not overlap. Hence the coverage of any set cover C equals the accumulated contribution of the corresponding selected subsets in the input domain of f , which completes the reduction.

□

N TRACTABILITY RESULTS FOR ORTHOGONAL DNFs

Lemma 14. *If f is an orthogonal DNFs and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time.*

Proof. We first define orthogonal DNFs (Crama & Hammer, 2011), then give the proof.

Definition 2 (Orthogonal DNFs). *A Boolean function φ is in orthogonal disjunctive normal form (orthogonal DNF) (Crama & Hammer, 2011) if it is a disjunction of terms $T_1 \vee T_2 \vee \dots \vee T_m$, such that for every pair of distinct terms T_i and T_j ($i \neq j$):*

$$T_i \wedge T_j \models \perp$$

Equivalently, no single variable assignment satisfies more than one term.

2160 The proof mirrors that of Theorem 1, reducing to the polynomial-time computability of v_{suff}^g for an
 2161 arbitrary set S .

2162 Fix a set S , and let $\mathcal{C}^S T_i$ denote the constraints that term T_i imposes on the variables in S . We
 2163 enumerate all term pairs (T_i, T_j) and discard any pair whose constraints on S are *inconsistent* (i.e.,
 2164 $\mathcal{C}_{T_i}^S \wedge \mathcal{C}_{T_j}^S \models \perp$), as it contributes zero to v_{suff}^g .

2166 For each *consistent* pair (T_i, T_j) , we evaluate $\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S)$, with $\mathcal{C}_{T_i, T_j}^S = \mathcal{C}_{T_i}^S \cap \mathcal{C}_{T_j}^S$; its
 2167 contribution to v_{suff}^g is

$$2168 \Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_i}^{\bar{S}}) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_j}^{\bar{S}}).$$

2171 Under feature independence, these probabilities factorize as $\Pr(\mathbf{x}_S) = \prod_{i \in S} p(x_i)$. Since there are at
 2172 most m^2 term pairs, v_{suff}^g is computable in polynomial time under an independent distribution $\mathcal{D}$. $\square$

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